FOREWORD

This is the second iteration of preparing our own courseware material, after successful completion of a similar task undertaken a few years ago. These contents have been carefully prepared and should serve as excellent auxiliary material for both instructors and students.

The Special Academic Group, Autonomy (SAGA) was formed for the sole purpose of preparing courseware contents primarily in the first-year theory subjects; a few second-year subjects were also included. The subjects for the first year were - Basic Electrical Engineering, Basic Electronics Engineering, Computer Programming, Data Structures and Algorithms, Engineering Mathematics I and II, Engineering Physics, Engineering Chemistry, Constitution of India, Environmental Science and Engineering and Communicative and Technical English. For the second year, the subjects for which courseware material was prepared were Analog Electronic Circuits, Digital Systems Design, Circuit Theory and Measurements and Instruments.
Faculty members from all the departments contributed to the task. They were, in no particular order, Nalini Singh, Bimal Meher, Saumyaranjan Dash, Mukti Routray, Susmita Biswal, Manasa Dash, Bipin Tripathy, Sibasankar Nayak, Janmejay Senapati, Subrat Sahu, Pradeep Moharana, Rupambika Pattanaik, Dhananjay Tripathy, Jagadish Patra, Sachin Das, Deepak Ranjan Nayak, Amulya Roul, Bodhisattva Dash, Sanghamitra Das, Gyana Ranjan Biswal, Nibedita Swain and Rajan Mishra.

The entire group worked diligently to successfully complete the task which included a peer review of the material. I take this opportunity to thank all the members of the SAGA group for a job well done.

I sincerely hope that this courseware material comes in handy and is utilized to the fullest extent. These are readily available additional resources prepared in accordance with the Silicon autonomy syllabus, to complement textbooks and classroom lectures. If there are any errors, I would be grateful if they are brought to my notice so that we can correct them in subsequent versions.

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Module – 1: ECOLOGY & BIOGEOCHEMICAL CYCLES

Historically, Rachel Carson's 1962 book "Silent Spring" started wide-scale awareness in the 60s about the environment; she battled some large chemical corporations and created public knowledge and understanding of the fragility of ecosystems, and the environment in general.

Origin

The word "Environment" has come from an old french word environ or environner to be exact which literally means "to encircle" or "to surround".

Definition

The word "environment" can be defined as "our life support system which includes air, water, land/soil & all other natural resources present around us".

According to ISO:14001 (International Organization for Standardization), environment can be defined as, "Surroundings in which an organization operates, including air, water, land, natural resources, flora(plants), fauna(animals), human beings and interrelations".

Related Terms

1. Environmental Science
2. Environmental Engineering
3. Environmental Management
4. Environmental Laws

1.Environmental Science

It is the study of human beings' impact on the environment and the physical, chemical and biological changes occurring in nature which focuses on pollution. It is highly interdisciplinary & it collects information from various fields such as biology, chemistry, geography, agriculture, geology etc.

2. Environmental Engineering

It is defined as that branch of engineering which is concerned with protecting the environment from the potentially deleterious or harmful effects of human activities, protecting the human population from the effects of environmental pollution and improving the environmental quality for better human health & well-being.

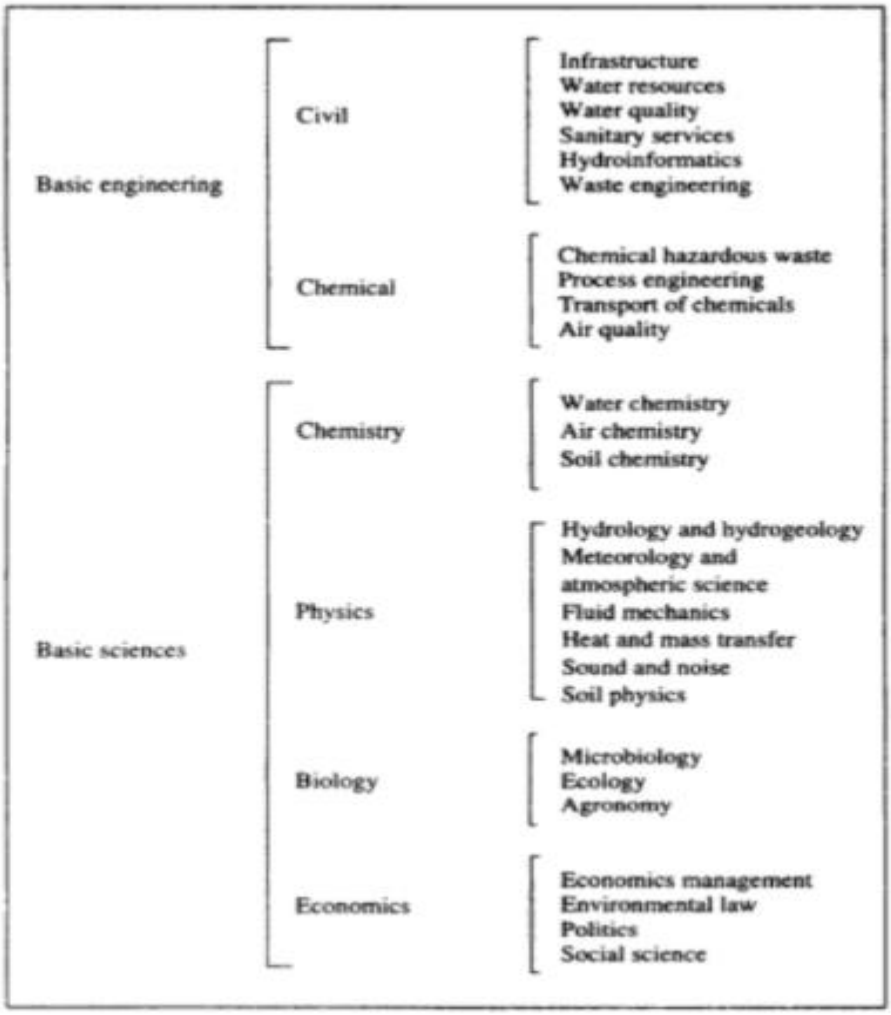
3. Environmental Management

It consists of auditing of EIA (Environmental Impact Assessment), Planning, Preparing EIS (Environmental Impact Statement) & pollution management.

4. Environmental Laws

It involves the body of statuaries, policies those are made in order to protect the nature from being misused by human beings. Some laws are regulative or controlling in nature while some are for preventive measure & some are binding on us.

Building Blocks of Environmental Engineering



Sustainable Development (SD)

It can be defined as "the ability to meet the needs of present generation without compromising the ability of future generation to meet their needs" i.e. the balance or proper use of resources, so that our future generation will not face any problem for their survival, after taking care of our own survival needs.

Pillars or Components of SD

There are three pillars or components of Sustainable Development (SD) i.e.

a) Economical Development

b) Society or community Development

c) Environmental Protection

Human Dimension in Env. Engg.

It is also known as Three factor model for assessing environmental degradation. To estimate or assess the extent of environmental degradation John Holdern (a physicist) and Paul Ehrlich (a biologist) has given a three factor model. According to this model, total environmental Impact (I) of degradation and pollution due to population in a given area depends on three factors i.e. P, A, T.

P represents Size of population

A represents Affluence or Per capita Consumption or per person consumption

T represents Technology used.

The relationship is, I = PAT

In the developing countries, **P^3 syndrome** i.e. (Population, Pollution, Poverty) and in the rich developed countries overuse of natural resources are the key or important factors leading to environmental degradation.

Environmental Problems

The major environmental problems that we are facing are –

a) Water Pollution

b) Air Pollution

c) Biodiversity Depletion

d) Large amount of Waste Production

e) Food Supply Problems

Causes of Environmental Problems

Many Environmental Problems presently we are facing are mainly due to –

a) Over Population

b) Wasteful use of Resources

c) Destruction & Degradation of Wildlife Habitats

d) Depletion & Contamination of Surface Water & Ground Water

e) Depletion of Non-Renewable resources or minerals

f) Deforestation

g) Soil Erosion

h) Loss of Biodiversity

Definition of Pollution

It is an undesirable change in physical, chemical or biological characteristics of air, water & soil, that may harmfully affect human beings, animals, the plant life, industrial progress, living conditions & cultural assets.

Definition of Pollutant

It is an undesirable harmful solid, liquid or gaseous substance present in such a concentration in the environment which tends to be injurious for the whole living organisms.

Ecological Concepts

The word ecology was coined by Ernst Haeckel from two Greek words, Oikos & Logos. Oikos means House or Living Space & Logos means Study. Ecology is the branch of Biology concerned with the relations of various species to one another & to their physical surroundings

Ecological Perspective

Ecological perspective demands a greater understanding of the functioning of living systems & their interactions with the environment. It gives a qualitative emphasis on ecological concepts.

Principles of ecology

- Living organisms can't exist with total disregard to their environment.
- All living organisms & their physical environment are interdependent upon each other & also affect each other.
- Every living organism has certain limits of tolerance towards the various factors of environment & only within these limits the organisms can survive.

- The existence of life depends upon flow of energy through food chains & the cycling of nutrients.

Aspects of ecology

The two important aspects of ecology are –

a) Autecology or species ecology

b) Synecology or community ecology

a) Autecology

It is concerned with the ecology of an individual species & it's population.

b) Synecology

It is the study of communities, their composition, their behaviour & the relation to the environment.

Synecology is further subdivided into three categories –

i) Population Ecology
ii) Genetic Ecology
iii) Taxonomic Ecology

i) Population Ecology

It relates with individual organisms with different groups of organisms within the ecosystem of different levels & the interrelationship among themselves.

ii) Genetic Ecology

It is also known as Gene Ecology. Different organisms have different genes and chromosomes; hence they live in different places.

The relationship due to the variation in genes among biotic components with their surroundings non-biotic environment is known as Genetic Ecology.

iii) Taxonomic Ecology

It includes ecology of taxonomic groups such as microbes, vertebrates (with backbone), invertebrates (without backbone) & insects etc.

Ecosystem

An ecosystem is defined as a natural functional ecological unit comprising of living organisms (i.e. biotic community) & their non-living (i.e. abiotic) environment that interact to form a stable self-supporting system.

Example: Pond ecosystem, Forest ecosystem etc.

The ecosystems are characterized by a diversity of species, but in an ecosystem there must be representatives from the three functional or metabolic groups such as Primary Producers, Consumers & Decomposers.

The ecosystems can vary in sizes.

Properties of Ecosystem

- An Ecosystem exists independent of specific components i.e. an individual tree may die, but a forest persist.
- The components of an ecosystem are interdependent.
- The nature of ecosystem depends on the species biodiversity of the ecosystem.
- The function of ecosystem depends on the energy flow & cycling of chemical elements within the ecosystems.
- The ecosystem can be disturbed by human activities & the most adverse effect of disturbance is the loss of biodiversity.

Types of Ecosystem

1. Natural Ecosystem
2. Artificial or Man-made or Man-engineered Ecosystem

1.Natural Ecosystem

These ecosystems operate by themselves under natural conditions.

Depending on the type of habitats, these are further sub-divided into –

I) Terrestrial Ecosystem

II) Aquatic Ecosystem

I) Terrestrial Ecosystem

It includes forest, grassland, desert ecosystems.

II) Aquatic Ecosystem

These ecosystems found in the aquatic environment.

It can be divided into two categories –

a) Fresh water ecosystem

b) Marine water ecosystem

a) Fresh water ecosystem

The Fresh water ecosystem can be

i) Lotic &

ii) Lentic

Lotic means (moving or running water)

e.g. River or Stream ecosystem

Lentic means (Still or Standing water)

e.g. Pond, Lake ecosystem

b) Marine water ecosystem

These include salt water bodies which may be Ocean ecosystem, Sea ecosystem.

2.Artificial ecosystem

These are maintained artificially by human beings.

e.g. garden, park.

Structure or components of ecosystem

Ecosystem has two major components.

1. Abiotic &
2. Biotic

1. Abiotic can be classified into hydrosphere, atmosphere & lithosphere
2. Biotic can be classified into producer, consumer & decomposer.

Consumer can be primary, secondary, tertiary consumers.

Functions of ecosystem

Energy flow: It regulates the flow of energy from one level to the other.

Nutrient cycling: It regulates cycling of nutrients.

Environmental gradients: The ecosystem fix the limit of tolerance for each organism towards various factors of environment.

Food Chain & Food Web: The energy produced by green plants are passed to next levels by a chain of consumers leading to formation of food chain & interlinking of food chain will lead to formation of food web.

Biodiversity: The ecosystem regulates the species diversity to acquire a stable system.

Natural Resources

The materials those occur in the environment or those are created by the environment & useful for supporting life or promoting the well-being of human beings are termed as natural resources

Natural Resources are the naturally occurring substances those are considered valuable in their natural form & their value rest in the amount of material available & the demand for it.

Types of Natural Resources

There are three types of natural resources –

1) Renewable or Inexhaustible

i.e. available in unlimited quantity.

Ex – Solar energy, wind power etc.

2) Non-renewable or exhaustible

i.e. available in limited quantity

Ex – coal, petrol, diesel etc.

3) Abstract Natural Resources

It includes animals, plants & natural landscapes as part of countryside used for recreation & tourism activities.

Ex – Bird watching, sight-seeing

Value of environment

The value of the environment is considered on the basis of the economic status of the particular region. Natural resources existing in the region are the backbone on the economic status.

The following four points will help in evaluating the environment –

1. Environmental Auditing
2. Biodiversity
3. Cost-Benefit Analysis
4. Environmental Ethics

1. Environmental Auditing

Environmental Audit is defined as a systematic listing of natural resources in an area playing a key role for economic status & sustainable developments of mankind in that area following definite management & regulatory systems.

Generally, it provides a list of all the resources present in the area.

The audit activities are carried out in three phases –

a) Pre-audit activities

b) On-site audit activities

c) Off-site or Post audit activities

1(a) Pre-audit activities

It includes planning, scheduling the audit activities & selecting the audit team.

1(b) Onsite audit activities

It includes interviewing of the site personnel & Inspecting the site.

1(c) Post audit activities

It includes –

- Preparing & distributing audit reports
- Determining potential solutions
- Assigning responsibilities
- Implementing actions &
- Conducting follow-up of the action plan.

2.Biodiversity

It doesn't imply just a collection of species, but is a basic resource which acts as a human life support system. Biodiversity plays an important role in biogeochemical cycles.

Biodiversity has three hierarchical levels i.e.

i) Genetic Biodiversity

ii) Species Biodiversity

iii) Ecosystem Biodiversity

There can be three kinds of losses of biological resources.

i) Depletion of a once common species

ii) Local or global extinction

iii) Ecosystem disruption.

3. Cost-Benefit Analysis

Working principles for Cost-Benefit Analysis include –

i) The SMS approach

SMS stands for Safe Minimum Standard

ii) The Minimax rule

Minimax stands for minimizing maximum losses

i) The SMS approach

It attempts to provide a decision-making process for problems involving long period of time & large uncertainties. This approach makes conservation the preferred option.

ii) The Minimax rule

It is a necessarily pessimistic decision rule biased towards conservation.

4. Environmental Ethics

It gives the attitude of people towards other living beings & the natural environment. We, human beings must eradicate our feelings of superiority over the rest of natural world & develop an "I-Thou" relationship rather than "I-It" relationship with the environment.

Levels of organization in Biotic Components

There are six major levels of ecological organization are recognized in the biotic components of the environment.

1) Individual

2) Population

3) Community

4) Ecosystem

5) Biomes &

6) Biosphere

1) Individual

These are single species. Individuals have physiological functions & respond to environmental conditions

2) Population

Population consists of a group of individuals of the same species living in a particular area at the same time. Birth rate, death rate plays an important role in the size of population.

3) Community

Population of different species living together interact with each other to form a community.

Niche

A suitable or comfortable position in the ecological hierarchy for a species.

Ex- Phytoplankton → Zooplankton → Fish → Whale

Niche of fish in the above ecological hierarchy is three or 3rd position.

4) Ecosystem

It includes both biotic & abiotic components of an area. The major or important feature of this ecological level is the strong interaction between the various biotic & abiotic components present. Nutrient cycling & energy flow occur in this ecological level

5) Biomes

When environmental conditions are similar in different parts of the country or around the globe, the habitats & communities are also often similar, giving rise to a higher level of biotic organization known as biomes.

Ex- a) **Tropical Rainforest** (High temp.& High rainfall)

b) **Desert Scrub** (High temp. & low rainfall)

6) Biosphere

The highest organizational level is the biosphere & it extends from bottom of the sea to the top of the sky where life exists. Biogeochemical cycles occur at this ecological level.

Ecosystem Process:

In the ecosystem process, the following processes are essential for the survival & maintenance of biotic components.

1. Energy flow &

2. Nutrient cycling

1. Energy flow

On the earth, the ultimate source of energy for life is solar radiation.

There are two sources of energy –

a) Autotrophic

b) Heterotrophic

1.(a) Autotrophic

Autotrophic production of energy is carried out within the ecosystem by green plants in the presence of sunlight using photosynthesis process.

1.(b) Heterotrophic

Heterotrophic energy source is the one, where the chemical energy is imported as organic matter which is originated from primary production in some other ecosystem.

This imported organic matter is called allochthonous.

Photosynthesis

$12H_2O + 6CO_2 + 709kcal$ (from sunlight) $\rightarrow C_6H_{12}O_6$(Carbohydrate)$+ 6O_2 + 6H_2O$

Out of total amount of solar radiation available, only 1 to 5% is used in the photosynthesis process. The organic matter produced by green plants in the Photosynthesis process is called Primary Production or (PP).

PP is affected by various environmental factors like water, light, temperature & soil nutrients. PP is of 2 types.

1) **GPP (Gross Primary Production)**

2) **NPP (Net Primary Production)**

GPP – R(Respiration) = NPP

GPP is the total amount of chemical energy or biomass stored by plants per unit area per unit time. Since plant requires energy for synthesis of organic matter & functioning of plant itself, some of GPP is used in the process of respiration.

NPP is used for plant growth & reproduction. NPP is normally 80 to 90% of GPP.

As various environmental factors affect PP, hence GPP & NPP vary over the globe. NPP can be classified into four broad groups each with a characteristic productivity range.

1. **Low range** – 0 to 250 gm/m^2 – year

   Ex- Desert, semi-desert

2. **Middle range** – 250 to 1000 gm/m^2 – year

   Ex- Non-forest communities like shrubland, grassland

3. **Normal range** – 1000 to 2000 gm/m^2 – year

   Ex- Forest

4. **High range** – 2000 to 3000 gm/m^2 – year

   Ex- Rainforest

Respiration

When any organism requires energy, the reverse chemical reaction of photosynthesis occurs, known as respiration, where the glucose molecule is broken down in the presence of oxygen to produce CO_2, H_2O & energy for work done & maintenance. The reaction is –

$C_6H_{12}O_6+6O_2 \rightarrow 6CO_2+6H_20+$ energy for work done & maintenance

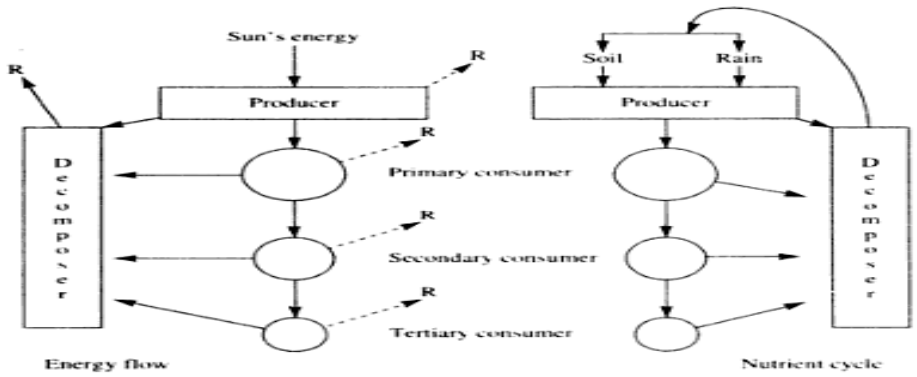
2. Nutrient cycling

During decomposition, the complex organic molecules in the original detritus or waste are gradually broken down to much simpler constituents & inorganic molecules like nitrates & phosphates, as the material moves through the decomposer or detritus food chain.

These are then, enter into the soil or sediment or dissolved in water, where they become the nutrients available for reuse by green plants.

This whole process of recycling of nutrients within the ecosystem is known as nutrient cycling.

Overview of 2 major ecosystem process of energy flow, nutrient cycling



Food Chain

Chemical energy produced by primary producers & the nutrients used by plants to build plant tissues, are passed up by a chain of consumers leading to formation of food chain, where each link in the chain is provided with energy & nutrients.

Depending on the types of foods, there are three types of food chain available.

a) Predator or Grazing Food Chain

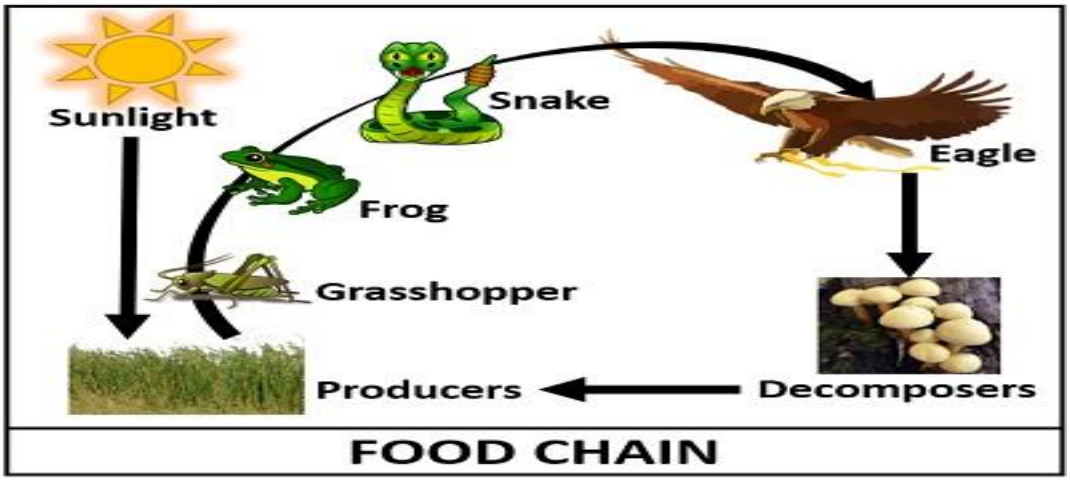
b) Saprophytic or Detritus Food Chain

c) Parasitic Food Chain

a) Predator or Grazing Food Chain

Predator means an animal that naturally preys on others.

Ex:- Sunlight→Grass→Grasshopper→Frog→Snake→ Hawk



b) Saprophytic or Detritus Food Chain

It starts from DOM (Dead Organic Matter) & goes to detritus feeding organisms i.e. detrivores i.e. bacteria, worms etc. & on to their predators.

Ex:- Dead leaves→Soil mites→Insects→Lizard

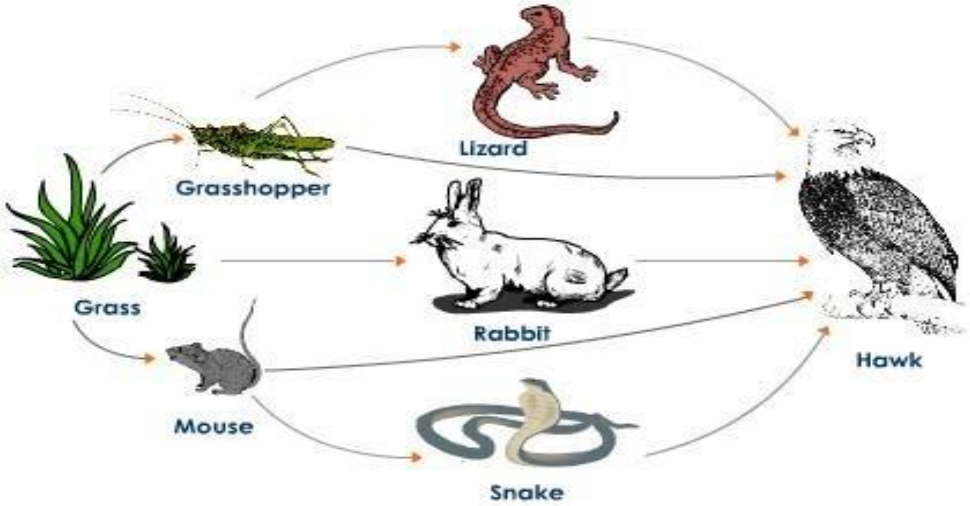
c) Parasitic Food Chain

It is a food chain in which parasites live on the host body or within the host body to get energy.

This food chain also starts from the green plants & animals to parasitic microbes.

Food Web

In most ecosystems, food chains interlink with each other to produce food web. Food web shows the food pattern of energy flow in the ecosystem.



A Food Web In a Grassland Ecosystem With Five Possible Food Chains

Tropic Level

Tropic level or nutritional level or tier gives the feeding status of an organism in an ecosystem. It is based on the concept who feeds on whom.

It can be -

i)**First tropic level or primary producer level**

Ex: - Green plants

ii)**Second tropic level or primary consumer level**

Ex: - Herbivores

iii)**Third tropic level or secondary consumer level**

Ex: - Carnivores

Ecological pyramid

The graphical representation of tropic levels is called the ecological pyramid.

The arrangement of organisms in a food chain according to tropic levels forms a pyramid.

Generally, pyramids are of three types.

1. Pyramid of energy

2. Pyramid of biomass

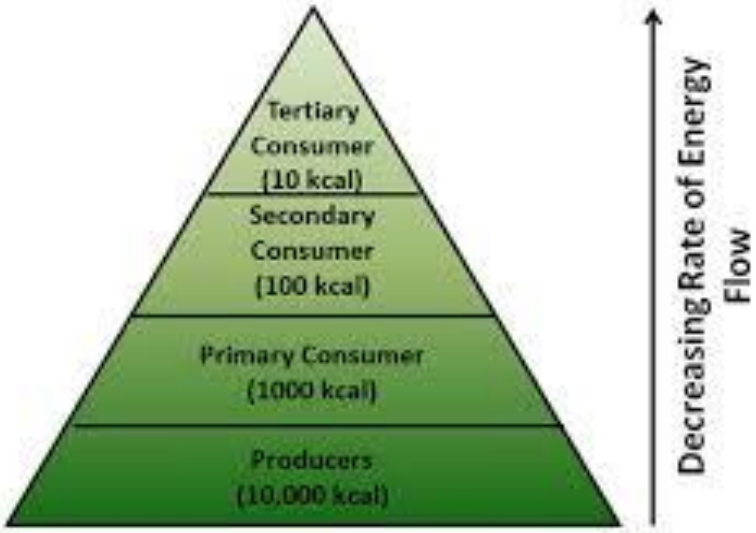
3. Pyramid of numbers.

1. Pyramid of energy

Generally, 10% energy is transferred between adjacent tropic levels & rest 90% of energy is used up in that tropic level giving rise to energy pyramid. This is called 10% rule of ecosystem.

It is based on rate of energy flow & productivity at each successive tropic level.

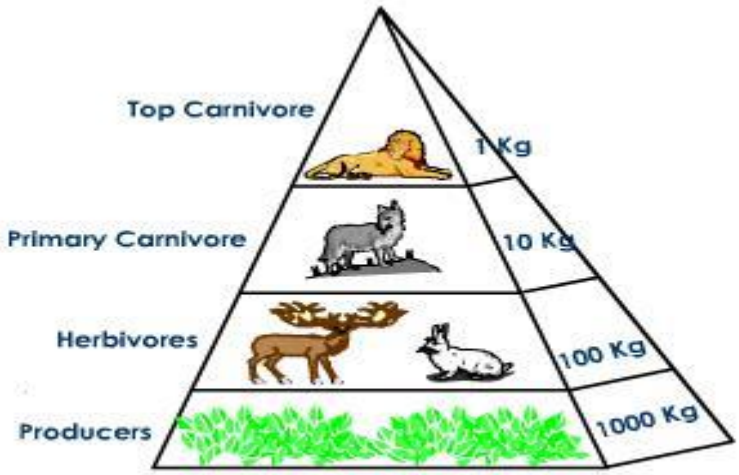
It decreases from autotrophs to higher tropic level.



Pyramid of Energy

2. Pyramid of biomass

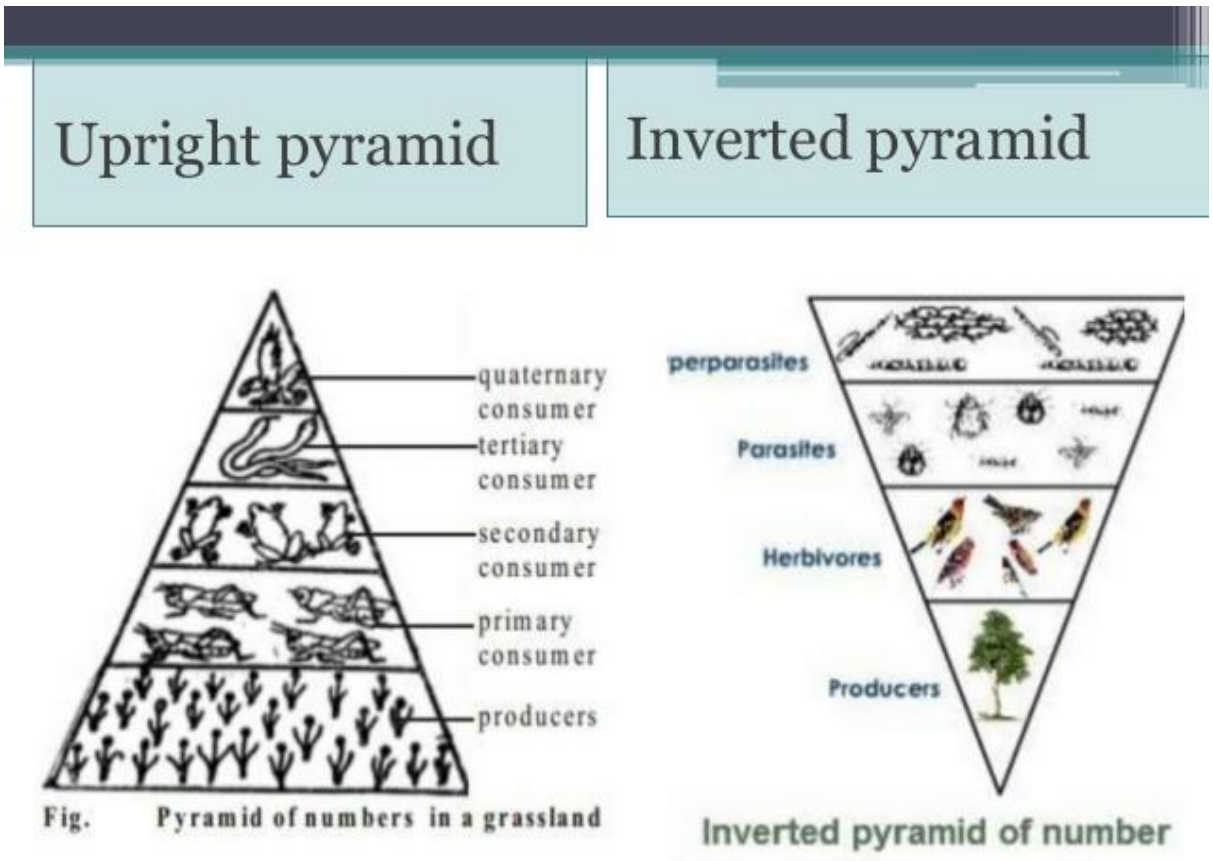
It shows the total mass of the organisms in each tropic level. It is based on total dry weight present in the level. It decreases from autotrophs to higher tropic level.



Upright Pyramid of biomass in a Terrestrial Ecosystem

3. Pyramid of Numbers

It shows the number of organisms in each tropic level. There may be gradual decrease or increase in the number of individuals.



Standing Crop

The amount of living materials in different tropic level or in any component population is known as **standing crop**. It can be expressed as number/unit area or as biomass.

Ecological Succession

In an ecosystem, new species may succeed older species. The process of gradual change in the conditions of physical environment, leading to the change in the species structure of an ecological community (i.e. replacement of one species by the other) over space & time, is called **Ecological Succession** or **Ecological Development.**

Biogeochemical Cycles

- Bio means living beings
- Geo means earth extended to air & water, where life exists.
- Chemical means chemical elements which continuously move in the cycles.

- The cyclic pathways through which chemical elements move from environment to the organisms & back to the environment are called bio-geochemical cycles.
- All parts or components of different ecological systems, on a local or global scale are linked with the Biogeochemical Cycles.
- Biogeochemical Cycles can have a number of phases & reservoirs.
  1) Organic Phase
  2) Inorganic Phase

1) Organic Phase

In this phase, nutrients pass rapidly through biotic communities using food chain.

2) Inorganic Phase

It contains all nutrient elements & are external to food chain.

The various sub-phases of the inorganic phase are –

a) Sedimentary Phase
b) Atmospheric Phase
c) Aquatic Phase or Aquatic Reservoir

a) Sedimentary Phase

It involves interactions with the solid earth or rocks & results of geological activities such as weathering.

b) Atmospheric Phase

It forms major part of some cycles like 'N' cycle & 'C' cycle & minor parts of some cycles like 'P' Cycle.

c) Aquatic Phase or Aquatic Reservoir

It involves plant nutrients.

Types of biogeochemical cycles

There are two types of biogeochemical cycles.

I) Air or Atmospheric Cycle.
II) Water or Hydrological Cycle

I) Air or Atmospheric Cycle

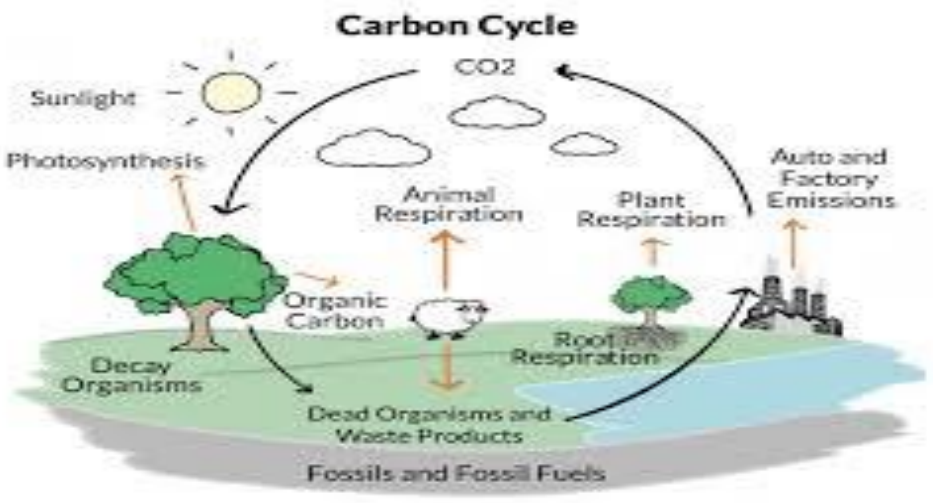
It can be classified into –

i) C – Carbon

ii) N – Nitrogen
iii) O – Oxygen
iv) P – Phosphorous
v)S – Sulphur

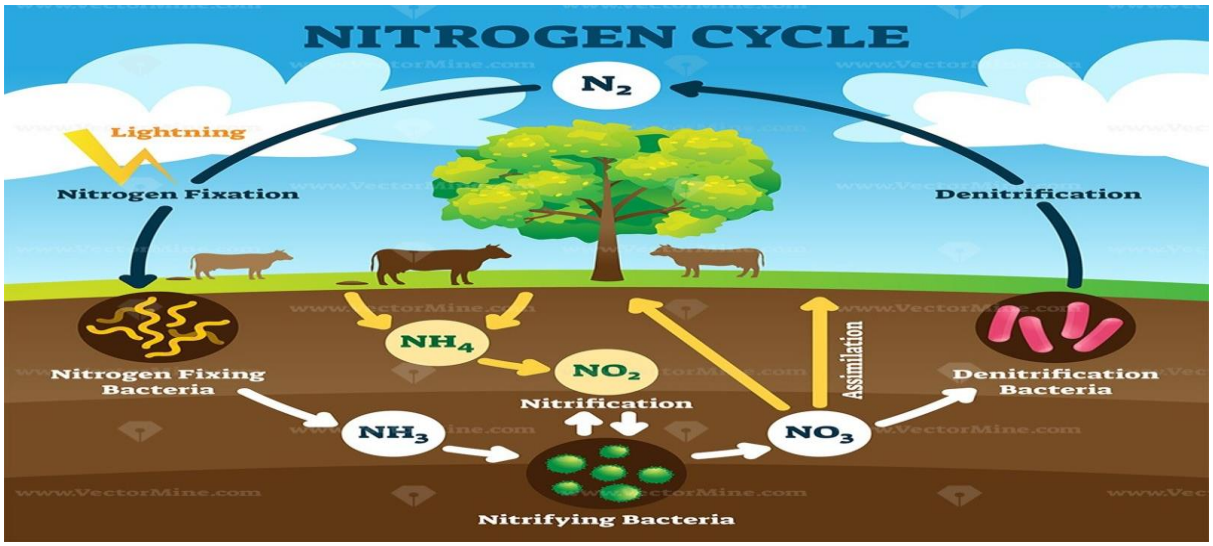
i) Carbon Cycle

The carbon cycle is nature's way of reusing carbon atoms, which travel from the atmosphere into organisms in the earth and then back into the atmosphere over and over again. Most carbon is stored in rocks and sediments, while the rest is stored in the ocean, atmosphere, and living organisms.



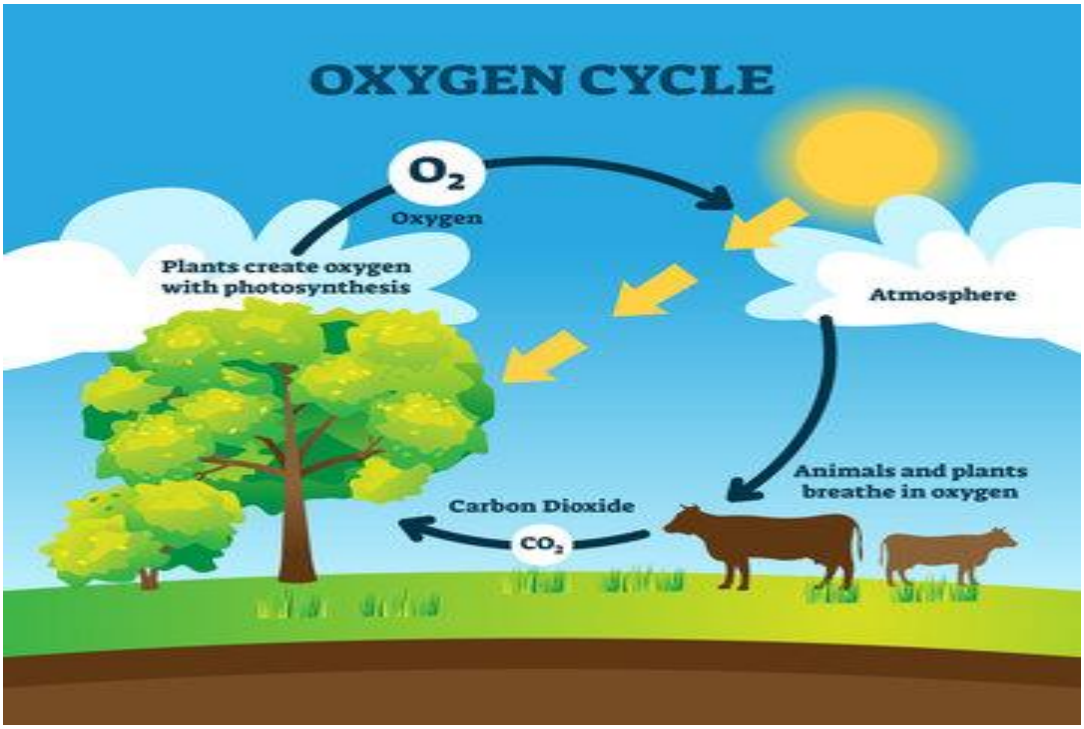
ii) Nitrogen Cycle

Nitrogen Cycle is a biogeochemical process through which nitrogen is converted into many forms, consecutively passing from the atmosphere to the soil to organism and back into the atmosphere. It involves several processes such as nitrogen fixation, nitrification, denitrification, decay and putrefaction.



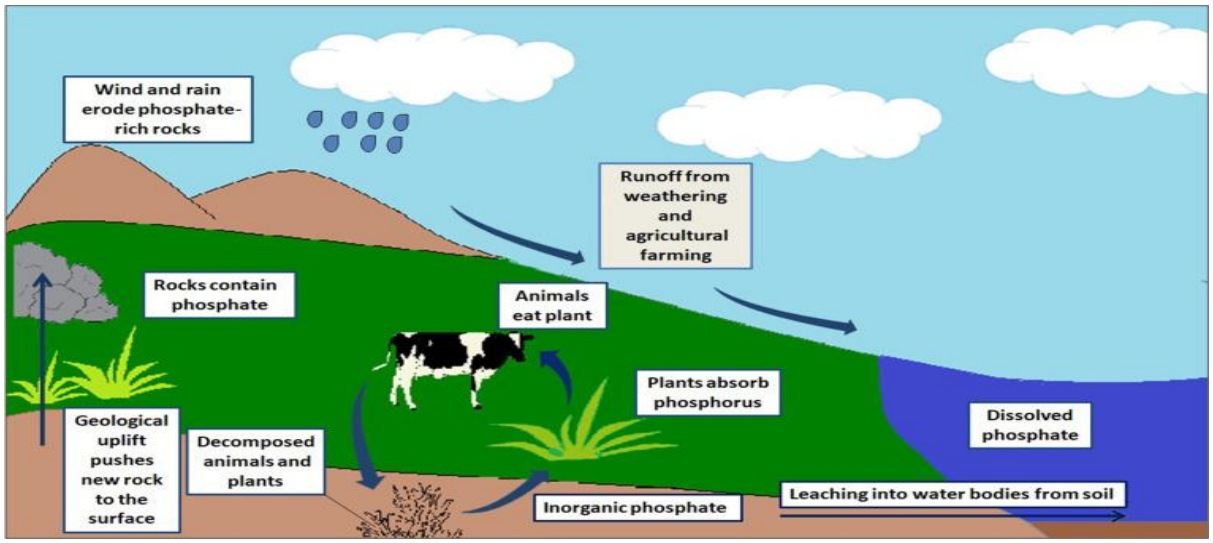
iii) Oxygen Cycle

The oxygen cycle is the cycle that helps move oxygen through the three main regions of the Earth, the Atmosphere, the Biosphere, and the Lithosphere. The Atmosphere is of course the region of gases that lies above the earth's surface and it is one of the largest reservoirs of free oxygen on earth.



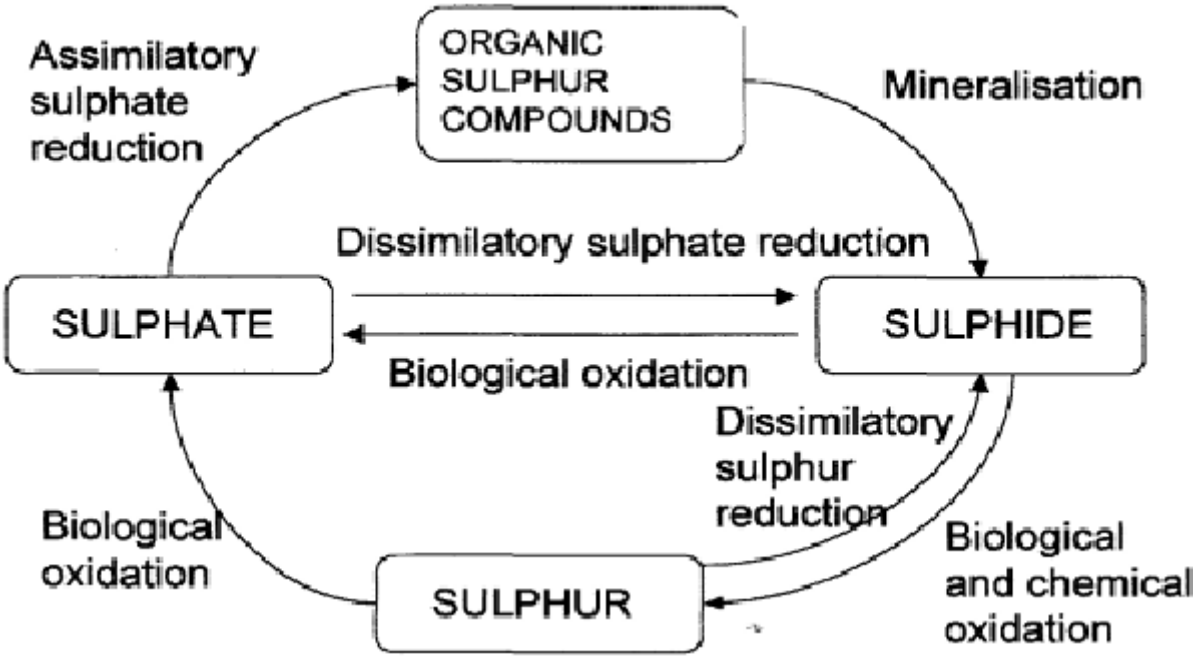
iv) Phosphorous Cycle

Phosphorus moves in a cycle through rocks, water, soil and sediments and organisms. Over time, rain and weathering cause rocks to release phosphate ions and other minerals. This inorganic phosphate is then distributed in soils and water. Plants take up inorganic phosphate from the soil.



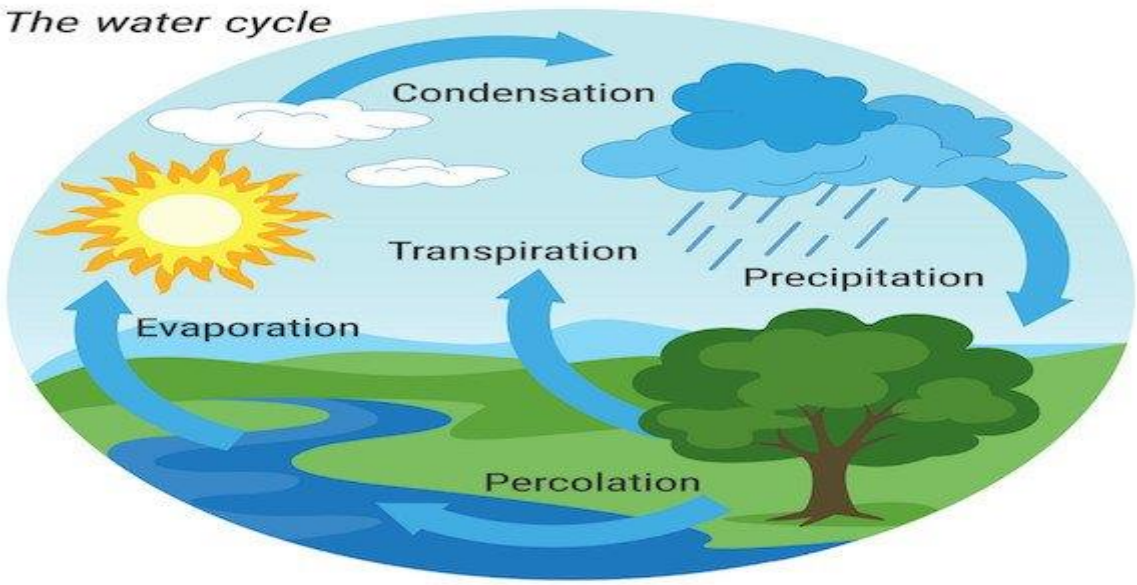
v) Sulphur Cycle

The sulphur cycle is the collection of processes by which sulphur moves between rocks, waterways and living systems. Such biogeochemical cycles are important in geology because they affect many minerals.



II) Water or Hydrological Cycle

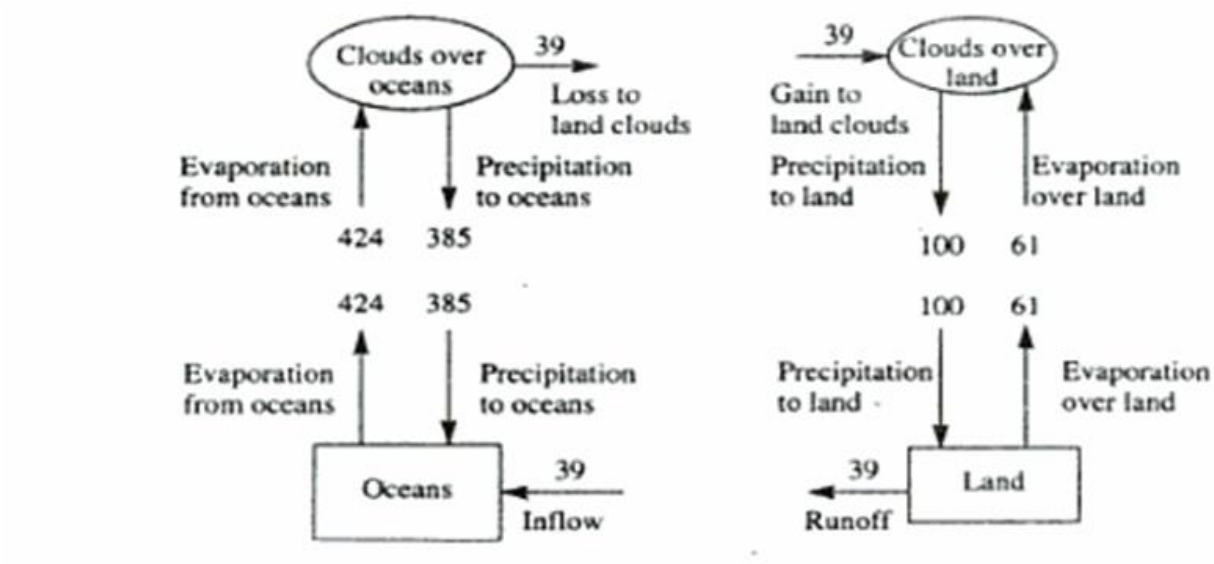
The hydrological cycle of the earth is the sum total of all processes in which water moves from the land and ocean surface to the atmosphere and back in form of precipitation. The hydrological cycle is dependent on various factors and is equally affected by oceans and land surfaces.



Interception:- It is the evaporation of water from the outer surface of leaves during & after rainfall.

Transpiration:- It is the evaporation of water through foliage.

Evapotranspiration:- It is the combination of Evaporation from water bodies & transpiration from leaves.



Material balance on aspects of the hydrological cycle.

Module – 2: WATER & WASTEWATER TREATMENT

Physical Properties of Water

The various physical properties of water that shows wide variation in magnitude includes –

- Colour
- Turbidity
- Taste & odour
- Temperature
- Solids
- **Colour**

It is measured in Hazen's Scale. The permissible limit of colour for drinking water is 5mg/L or ppm.

- **Turbidity**

It is the measure of cloudiness in water & caused by presence of suspended matter which scatters & absorbs light. It is measured in NTU (Nephelometric Turbidity Unit). The Permissible limit of turbidity for drinking water is 1NTU.

- **Taste & Odour**

This may be due to presence of micro-organisms, algae, dissolved minerals, salts etc. It is measured in TON (Threshold Odour Number). The permissible limit of TON for drinking water is 1 to 3.

TON = (A + B)/ A

A - Volume of Sample with odour
B -Volume of Pure Water with no odour Added
If A was a 100 ml sample and 100 ml of water had to be added to not detect the odour - the TON would be 2.

TON = (100 + 100)/ 100=200/100=2.

- **Temperature**

The temperature of water varies with atmospheric temperature & it is a significant parameter regarding characteristics of water. For drinking water, the temperature should be 10-25^{0}C.

- **Solids**

Solids content of water represents the characteristics of water.

Solids present in water can be studied as follows-

a) Total Solids (TS)
b) Suspended Solids (SS)
c) Total Dissolved Solids (TDS) = TS – SS
d) Total Volatile Solids (TVS)
e) Volatile Suspended Solids (VSS)
f) Filterable Solids (FS) & Non-Filterable Solids (NFS)

NFS = SS (Particle Size Range is 10^0 to 10^{-3} mm)

FS can be Colloidal (Particle Size Range is 10^{-3} to 10^{-6} mm) &

Dissolved (Particle Size Range is 10^{-6} to 10^{-9} mm)

Chemical Properties of Water

It can be Inorganic & Organic

Inorganic Chemical properties of water includes

- P^H
- Alkalinity & Acidity
- Hardness
  Conductivity

Organic Chemical properties of water includes

- BOD (Biological Oxygen Demand)
- COD (Chemical Oxygen Demand)
- TOC (Total Organic Carbon)
- TOD (Total Oxygen Demand)

Inorganic Chemical properties of water

- **P^H**

P^H is Potential of Hydrogen or Power of Hydrogen

$$P^H = -\log [H^+] \text{ or}$$
$$= -\log [H_3O^+]$$

P^H range for public water supplies is 6.5 to 8.5.

Q. In a water treatment plant, the P^H values of incoming & outgoing water are 7.2 & 8.4 respectively. Assuming a linear variation of PH with time, determine the average P^H value of water.

- **Alkalinity & Acidity**

Alkalinity is also known as ANC or (Acid Neutralizing Capacity) & Acidity is also known as BNC or (Base Neutralizing Capacity)

Alkalinity can be of three types.

a) Carbonate Alkalinity

b) Bicarbonate Alkalinity &

c) Hydroxide Alkalinity

- **Hardness**

Hard water has high mineral content. Hardness of water are of two types.

a) Temporary Hardness or Carbonate Hardness

b) Permanent Hardness or Non-Carbonate Hardness

a) Temporary Hardness

It is due to carbonate & bicarbonate of Ca^{+2} & Mg^{+2}.

b) Permanent Hardness

It is due to Chloride & Sulphate of Ca^{+2} & Mg^{+2}.

Hardness of water is expressed in mg/L as $CaCO_3$. The permissible limit of hardness of drinking water is 200 mg/L as $CaCO_3$.

Hardness in mg/L as $CaCO_3$

= [M^{+2}(concentration in mg/L) X 50]/ [equivalent weight of M^{+2}]

= M^{+2}(Concentration in meq/L) X 50

Q. Determine the various hardness of the following water sample.

Constituent	Concentration (in mg/L)
Ca^{+2}	60
Mg^{+2}	29.3
HCO_3^-	366

- **Conductivity or Electrical Coductivity**

It is the measure of the ability of an aqueous solution to carry an electric current. The equation can be,

$ec = \Sigma C_i f_i$

where,

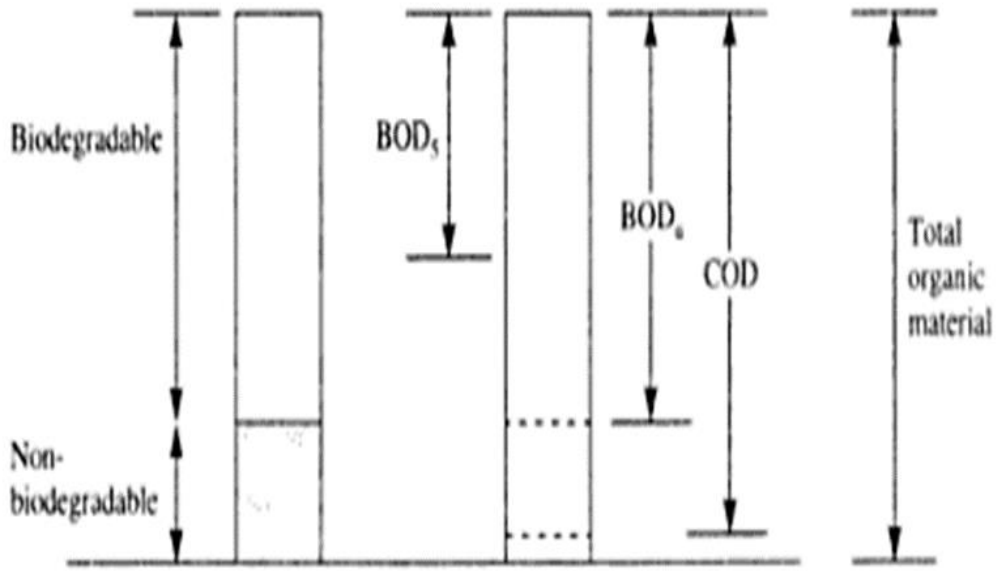
ec is electrical conductivity in µs(siemens)/cm,

C_i is concentration of ionic species in solution in mg/L &

f_i is conductivity factor for ionic species.

Organic Chemical properties of water

- **BOD_5 = 0.6 COD**
- **CBOD or Carbonaceous BOD = 0.92 COD**



Schematic of BOD/COD/T.org matter relationship.

Q. Determine COD & TOC for glucose($C_6H_{12}O_6$)?

Q. If bacteria cells are represented by the chemical formula $C_5H_7O_2N$, then determine the potential carbonaceous BOD, when bacteria cell concentration is 1000mg/L.

Q. A wastewater is analysed & is shown to contain 100 mg/L of ethylene glycol ($C_2H_6O_2$) & 120 mg/L of phenol(C_6H_6O). Determine COD & TOC.

Organic Content Parameter of Water

When an organic waste is discharged to a stream, the organic content of the effluent or discharge undergoes a biochemical reaction with the help of micro-organisms.

$BOD = P(DO_I - DO_F)$

Where, P is Dilution Ratio,

DO_I & DO_F are initial & final DO (Dissolved Oxygen) concentration

Corrected BOD = $P[(DO_I-DO_F)-f(B_I-B_F)]$

Where, f is the correction factor or seeding factor=1-(1/P),

B_I & B_F are initial & final DO (Dissolved Oxygen) concentration of seeded diluted water or Blank

Q. The results from a BOD test diluted by 100 are given below. Calculate the BOD_S.

Time (in days)	DO(in mg/L)
0	7.95
1	3.75
2	3.45
3	2.75
4	2.15
5	1.80

Q. The results from a BOD test diluted by 100 are given below. Calculate the Corrected BOD_S.

Time (in days)	DO(in mg/L)	Blank DO(in mg/L)
0	7.95	8.15
1	3.75	8.10
2	3.45	8.05
3	2.75	8.00
4	2.15	7.95
5	1.80	7.90

The rate of decomposition of organic matter is directly proportional to the amount of organic matter available & it is a first order reaction.

i.e. $d/dt(L_t) \alpha L_t$

=> $d/dt(L_t) = -k_1 x L_t$

=> $d(L_t)/L_t = k_1 x dt$

Where, L_t is BOD remaining in mg/L=BOD_r &

K_1 is deoxygenation rate coefficient

Now taking integration of both sides and putting initial condition at t=0, $L_t=L_0$, we will have final expression for BOD exerted at any time t i.e.

$BOD_t = L_0(1-e^{-k_1 t})$

Q. If K_1 or deoxygenation rate coefficient is 0.15 per day, then determine the ultimate BOD or BOD_u or L_0.

Temperature effect on K_1

Temperature has an effect on K_1 & is given by,

$$K_T = (K_{20}) x (\Theta)^{(T-20)}$$

Where,

K_T= The deoxygenation rate coefficient at T^0C

K_{20}=The deoxygenation rate coefficient at 20^0C

Θ = Coefficient= 1.047 for $20^0C < T < 30^0C$ &

= 1.35 for $4^0C < T < 20^0C$

NBOD

The Oxygen demand associated with the oxidation of ammonia (NH_3) to nitrate (NO_3^-) is called Nitrogenous BOD or NBOD.

$2NH_3 + 3O_2 \rightarrow 2NO_2^- + 2H^+ + 2H_2O$

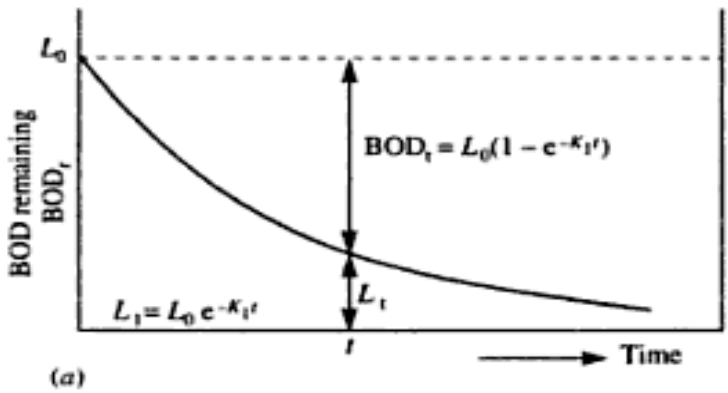
$2NO_2^- + O_2 \rightarrow 2NO_3^-$

So $2NH_3 + 4O_2 \rightarrow 2NO_3^- + 2H^+ + 2H_2O$

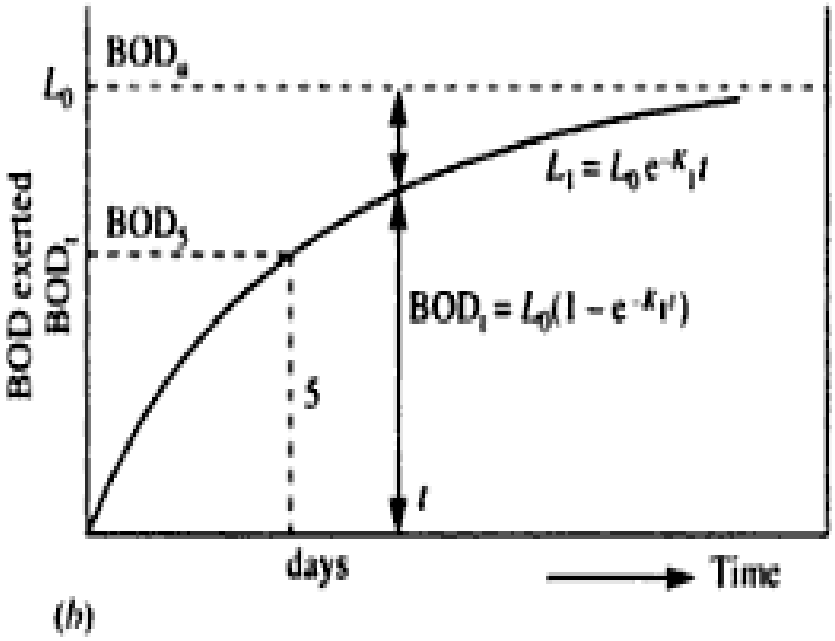
Finally, $NH_3 + 2O_2 \rightarrow NO_3^- + H^+ + H_2O$

Q. Determine the TOD (Total Oxygen Demand) if the sample contains 25 mg/L of N.

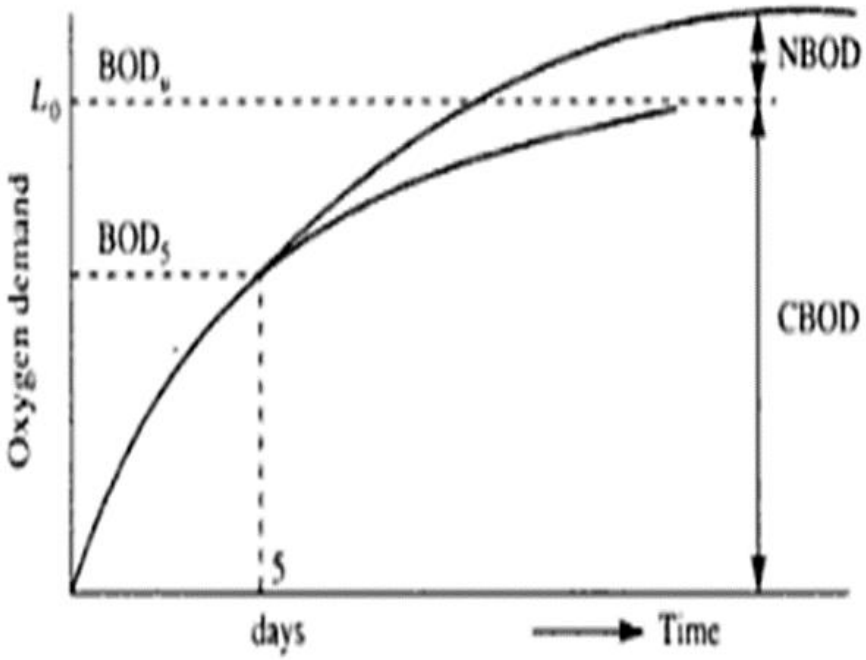
BOD remaining (BOD_r) graph



BOD exerted (BOD_t) graph



CBOD & NBOD relationship



Relationship of CBOD and NBOD.

Q. 15ml of sample was diluted in 300ml bottle. Initial DO was 8.9 mg/L & final DO after 5days was 4.4mg/L. The corresponding initial & final DO of diluted water was 9.1mg/L & 9.05 mg/L. Find out 5 days BOD at 20^0C.

Q. BOD of an sample incubated for one day at 30^0C was found to be 100mg/L. What would be the 5 day BOD at 20^0C. The deoxygenation rate coefficient (K_1) is 0.12 at 20^0C.

Q. Determine the two day BOD for a wastewater at 20^0C, while BOD_5 at 20^0C is 250mg/L. The deoxygenation rate coefficient(K_1) value at 20^0C is 0.2 per day. What shall be the BOD after the end of seven days at 25^0C .

Q. A BOD test was conducted taking 5% wastewater mixed with 95% aerated water for dilution & the following observations were found out:

i) DO of the aerated water used for dilution is 3.6 mg/L

ii) DO of the original wastewater sample is 0.8 mg/L

iii) DO of the diluted wastewater after incubation at 20^0C for 5 days is 0.7 mg/L

Calculate the 5 days BOD of the above wastewater sample if the deoxygenation constant is 0.11.

Aquatic Pollution

The aquatic Pollution can be classified into three broad groups.

a) Fresh water pollution or Surface water pollution

b) Estuarine pollution

c) Marine pollution

a) Fresh water pollution

The pollution of fresh water can be caused by:-

- Wastes from municipality & industries
- Oil from industry wastes & cleaning of vehicles
- Thermal waste from industries & power stations
- Heavy metals etc.

b) Estuarine pollution

Estuarine means tidal mouth of river or where river mixes with the sea. So if surface water is polluted then it can lead to pollution of estuarine.

c) Marine pollution

The main pollutants for Marine pollution are:-

i) Manmade organic poisons like Pesticides & herbicides

ii) Radioactivity due to radioactive substances.

Point & Non-Point Sources of Pollution

Point Sources are those, which discharge pollutants from specific locations such as chimney, drainpipe of factories etc.

Point Sources are easier to monitor & regulate.

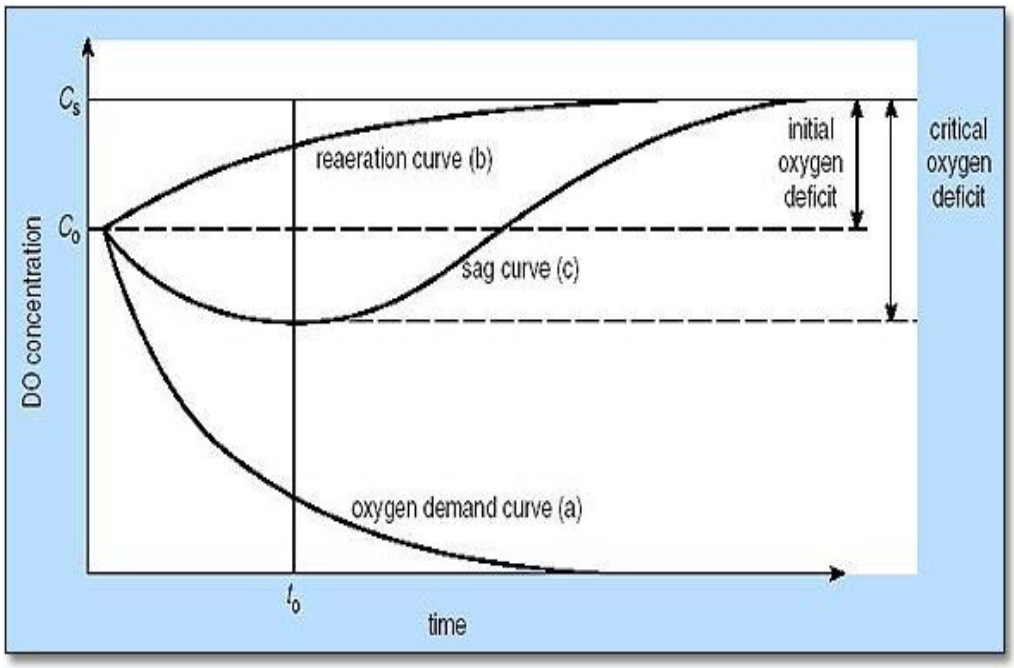
Non-point sources of pollution are those, which don't have any specific location of discharge.

As a result, these are difficult to monitor & regulate.

Ex:- Run-off from paddy fields, Lawns & Gardens.

Oxygen Sag Curve

The longitudinal profile of Oxygen concentration is Known as Oxygen sag curve.



Eutrophication

It can be defined as the enrichment of water by inorganic plant nutrients like N & P, which results in increase in primary productivity.

Artificial eutrophication is known as cultural eutrophication which is caused by adding fertilizers.

The problems arising due to Eutrophication are:-

- Water quality may have unacceptable taste & odour
- Water may be injurious to health

- The household usefulness of water may decrease.

Oxygen transfer in waterbodies

The oxygen transfer in waterbodies is given by Orlob equation i.e.

$\{(C_g-C_0)/(C_g-C_t)\} = e^{k_2}t$

Where,

C_g = Saturation Concentration of oxygen

C_0 = Initial Concentration of oxygen

C_t = Concentration of oxygen after time 't'

K_2 = reaeration rate constant

Q. If saturation concentration of oxygen in water is 9.2mg/L & the initial concentration of oxygen is 2.1mg/L, then determine the time it takes to become 7.5mg/L, having K_2 value 0.25 per day. What is the time required for C_t to be 5,6,7,8,9 mg/L.

Ground water Quality/Contamination

The simplest model of decay of a contaminant is to assume the decay to be a first order reaction i.e.

$r = - kc$ &

$C_t = C_0e^{-Kt}$

Where,

C_t = Concentration at time 't' in mg/L

C_0 = Initial Concentration in mg/L

t = time in days

K = first order decay coefficient

Q. Determine the concentration of a contaminant at the downstream well, if the upstream concentration is 80mg/L. Assume decay constant (k) to be 10^{-4} per day & time (t) = 274 years

Ground Water Recharge

It is also known as deep percolation. It is a hydrologic process, where water moves downward from surface to ground.

Recharge can occur both naturally & artificially.

- Natural way of recharge is occurred through hydrological cycle.
- Recharge occurs artificially, when rainwater, reclaimed water or recycled water is injected or routed to the subsurface.

Recharge can help move excess salts that accumulate in the root zone to deeper soil layers or into the deep groundwater system.

Ground Water Flow

The ground water flow is governed by Darcy's law which can be expressed as follows:-

$Q = KiA$

$= KA(dh/dL)$

$= KA\{(h_2-h_1)/(L_2-L_1)\}$

Where,

Q = horizontal flow in m^3/sec

K = hydraulic conductivity in m/sec

i = dh/dL= hydraulic gradient

A = cross sectional area in square mtr

h_2- h_1= water head drop

L_2-L_1 = Length difference over which water head drop occurs

q = Q/A = Specific discharge or discharge per unit area

Q. Determine the daily flow capacity and transmissivity of an aquifer, if it's height, width, length are 15mtr, 800mtr & 2km respectively. The water head change over the length 2km is 3mtr. The hydraulic conductivity (K) is $6x10^{-7}$ m/sec.

Water Table

A water table is the underground depth, at which point the ground is totally saturated by water. The water table may vary due to seasonal changes in precipitation, evapotranspiration & topography etc.

Aquifer

An aquifer is a body of saturated rock through which water can easily move. An aquifer is an underground layer of water bearing permeable rock from which ground water can be extracted using a water well.

Aquifer can be of two types-

- a) Unconfined aquifer
- b) Confined aquifer

a) Unconfined aquifer

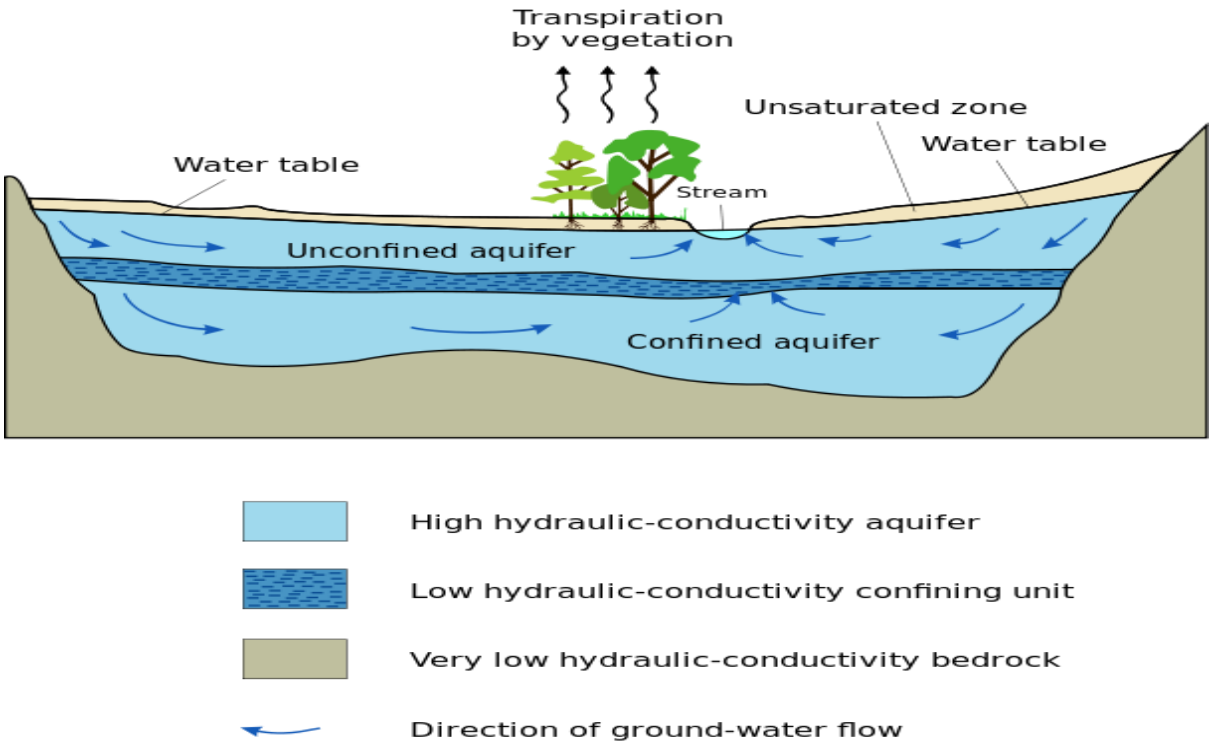
An unconfined aquifer is the one which is open to receive water from the ground surface & whose water table surface is free to fluctuate up & down, depending on the recharge & discharge rate.

These are extremely susceptible to contamination.

b) Confined aquifer

These types of aquifers have rock layer with less permeability or confining bed of rock above them due to which water can't enter freely into these type of aquifers from the ground surface.

These are less susceptible to contamination.



Concept of Hydrology

i) Water Balance or Water Budget

ii) Hyetograph & Hydrograph

iii) Rainfall-Runoff Relationship or Rational formula for determining runoff

i)Water Balance or Water Budget

It gives the quantitative account of hydrological cycle. The equation for water balance is based on conservation of mass.

The water balance equation is

$P = R + E \pm \Delta S \pm \Delta G$

Where,

P is Precipitation

R is Runoff

E is Evaporation

ΔS is change in soil moisture status

ΔG is change in ground water status

ii) Hyetograph & Hydrograph

Hyetograph represents rainfall intensity vs time.

Hydrograph represents runoff or stream discharge vs time.

If time is unit i.e. 1hr., then hydrograph will be unit hydrograph.

iii)Rainfall-Runoff Relationship or

Rational formula for determining runoff

$Q = CIA$

Where,

Q is runoff in m^3/sec

C is locality coefficient

I is Rainfall Intensity in mm/hr

A is Catchment Area in square KM

Peak Discharge (QP)or Peak Run-off

It can be expressed as follows-

$Q_P = 0.278\ CIA$

Modified Rational Formula given by Wallingford is-

$Q_P = 0.278C_VC_RIA$

Where, C_V is volumetric runoff coefficient & C_R is Routing Coefficient

Water Treatment

The objectives of water treatment are-

A) To produce Water which is safe for human consumption

B) To produce Water at a reasonable cost

C) To reduce water impurities to acceptable levels

Organoleptic Parameters

Organoleptic Parameters are those parameters which can be sensed or felt by human organs.

Ex:- Colour, Taste & Odour etc.

Classes of Water treatment

Class	Descripition	Source
A	No Treatment	Borehole water
B	Disinfection only	Borehole water
C	Standard water Treatment	River,Reservoir
D	Advanced/Special water Treatment	Industrial Water

Monitoring Frequency

The water quality parameters are required to be monitored depending on the source & quality of raw water.

The three categories are-

A) Minimum monitoring(C1)

B) Current monitoring(C2)

C) Periodic Monitoring(C3)

Water Treatment Processes

The treatment processes of water can be classified into three stages-

A) Pretreatment processes

B) Standard or Conventional processes

C) Special or Advanced processes

A) Pretreatment processes

Some treatment processes are required prior to standard water treatment processes which may include-

i) Screening

ii) Storage

iii) Aeration

iv) Chemical pre-treatment

i) Screening:-

Different types of screens are used in this process. Those are can be -

a) Coarse Screen

b) Fine Screen

c) Micro Screen

a) Coarse Screen:-

These screens contain typically inclined bars of 25mm dia & 100mm spacing, preventing large floating matters from entering the treatment plant

b) Fine Screen:-

If storage is not provided, then fine screens are fitted after the coarse screens. If there is storage, then fine screens are placed at the outlet of the storage tanks.

Fine screens are typically mesh with openings about 6mm dia or square

C) Micro Screen:-

In micro screens, mesh openings range from 20 to 40 µm.These type of screens are used only for relatively uncontaminated water.

ii) Storage:-

Storage tanks balance the flows going to the treatment plant. Storage capacity should be equivalent to 7 to 10 days of the average water demand. The period of storage shouldn't be long, so as not to encourage the growth of unwanted organisms

iii) Aeration:-

Aeration is the supply of oxygen from the atmosphere to water to effect beneficial changes in the quality of water. It is a common treatment process for groundwater

Aeration is generally used for the followings-

a) To release excess H_2S gas which may cause undesirable taste & odour.

b) To release excess CO_2 which may have corrosive properties

c) To increase oxygen content of water.

iv) Chemical pre-treatment:-

The chemical pre-treatment to remove undesirable properties of water like algae or excess colour is a more expensive process than chemical post-treatment. In pre-treatment, greater amount of chemicals are required to effect the same result as some of the chemicals are absorbed by turbidity of water.

The two Chemical pre-treatment processes which are generally used are-

a) Pre-Chlorination

b) Activated Carbon

a) Pre-chlorination:-

It is used on low turbidity water with a high coliform count. The chlorine is injected into the water stream & it oxidizes & precipitates iron & manganese. It also kill pathogens & reduces colour. The chlorine dose used is 5mg/L. Pre-chlorination also reduces NH_3 in both surface water & ground water supplies.

b) Activated Carbon:-

The activated Carbon is used for-

i) Removal of Photosynthetic Algae

ii) Improvement of colour & odour

iii) Removal of organic compounds

Activated Carbon can be used either as PAC (Powdered Activated Carbon) or as GAC (Granular Activated Carbon). Generally for water treatment PAC is used, but GAC is used where taste & odour of water have an industrial base.

PAC has lower cost & efficiency than GAC. Doses may vary from 3 to 20 mg/L.

B) Standard or Conventional processes

It includes-

i) Sedimentation

ii) Coagulation

iii) Flocculation

iv) Filtration &

V) Disinfection

i) Sedimentation

The heavier large particles settle down by the force of gravity at the bottom of the sedimentation tank, which is called Sedimentation. Stoke's law for settling velocity is the deciding force for sedimentation.

Stoke's law can be represented by-

$$V_S=(g/18\mu)(\rho_s-\rho_w)d^2$$

Where,

V_S is settling velocity

g is acceleration due to gravity

μ is dynamic viscocity

d is dia of settling particle

ρ_s is density of particle settling

ρ_w is density of water

ii) Coagulation

The lighter smaller particles don't settle down by the force of gravity. To make them settle down, some amount of coagulants are added to the water, which is called Coagulation.

Some examples of coagulants are-

- Alum i.e. aluminium sulphate
- Ferric Chloride
- Ferric Sulphate
- Ferrous Sulphate

iii) Flocculation

When coagulants are added to the water, the lighter smaller particles coagulate or combine with each other forming bigger particles, known as floc & the process is called flocculation.

iv) Filtration

It is the process of passing water through a porous medium, with the expectation that the filtrate water has better quality than the incoming water. Generally, two types of filters are used

a)**Rapid gravity filter**:-

Here rate of filtration is high i.e. 5 to 20m/hr.

b) **Slow sand filter**:-

Here rate of filtration is slow i.e. 0.1 to 0.2m/hr.

Criteria	Rapid sand filters	Slow sand filters
Improvement of water quality	With pre-treated raw water, a filtrate quality is possible that has less than 1 NTU, 90% removal of coliforms, 50 – 90% removal of Cryptosporidium and Giardia cysts, 10% removal of colour, 5% removal of Total Organic Content (WEDC, 1999 [ref 02]). As a single process, rapid sand filtration was ranked third most effective of all treatment processes, given a range of pathogenic, chemical and aesthetic factors.	With raw water, a filtrate quality is possible that has less than 1 NTU, 95% removal of coliforms, 99% removal of Cryptosporidium and Giardia cysts, 75% removal of colour, 10% removal of Total Organic Content (WEDC, 1999 [ref 02]). As a single process, slow sand filtration was ranked second most effective of all treatment processes, given a range of pathogenic, chemical and aesthetic factors.
Rate of flow	A flow of between 4 – 21 m/h can be expected from a rapid sand filter, which is somewhere between 20 and 50 times faster than the range of slow sand filtration.	Flow rates are usually around 0.1 m/h but can increase up to 0.4 m/h. Check out Flow Rates for more information.
Filter media	Rapid sand filters are made using graded sand, sometimes with an additional coarser layer of material on top of the sand to increase the flow rate (for example, anthracite), in which case they become known as dual-media filters. The effective size for rapid filters is usually greater than 0.55mm with a Uniformity Coefficient of less than 1.5.	Slow sand filters on the other hand, should ideally have an effective size of between 0.15 – 0.35mm, and a Uniformity Coefficient of between 1.5 – 3, but preferably less than 2.
Penetration of solid matter	Penetration of suspended matter into the sand bed is deeper for rapid sand filters, which are usually cleaned by backwashing.	Solids only tend to penetrate slow sand filter beds by between 0.5 and 2 cm by comparison, allowing more manual methods to be employed for cleaning.
Pre-treatment	Pre-treatment is usually necessary for rapid sand filtration. Such treatment could include coagulation and flocculation, followed by sedimentation.	No pre-treatment is usually necessary for raw waters with turbidities of less than 50 NTU. Huisman and Wood (1974 [ref 03]) suggest that the best filtration occurs when turbidity is less than 10 NTU. Note that coagulation and flocculation are not appropriate pre-treatments for slow sand filters because floc carryover is possible, which rapidly blocks the sand bed.

V) Disinfection

By disinfection, pathogens & micro-organisms are killed, thereby making water more pure.

The requirements of a good disinfectant are-

a) It should be toxic to micro-organisms.

b) It should have a fast rate of kill.

c) It should be persistent enough to prevent regrowth of organisms in the distribution system.

d) It shouldn't produce undesirable compounds.

e) It should be safe to handle.

f) It should be of reasonable cost.

Chick's law

The rate of destruction of micro-organisms is often a first order chemical reaction as given by Chick's law i.e.

$$\frac{d}{dt}(N_t)\alpha(N_t) \Rightarrow \frac{d}{dt}\left(N_t\right) = -k(N_t)$$

$$So, N_t = N_0 e^{-kt}$$

Where,

N_t = Number of organisms at time 't'

N_0 = Number of organisms at time 't'= 0

K = Rate constant depends on types of micro-organisms present & types of disinfectants used

The various examples of disinfectants are-

a) Chlorine dioxide

b) Chlorination

c) Chloramines

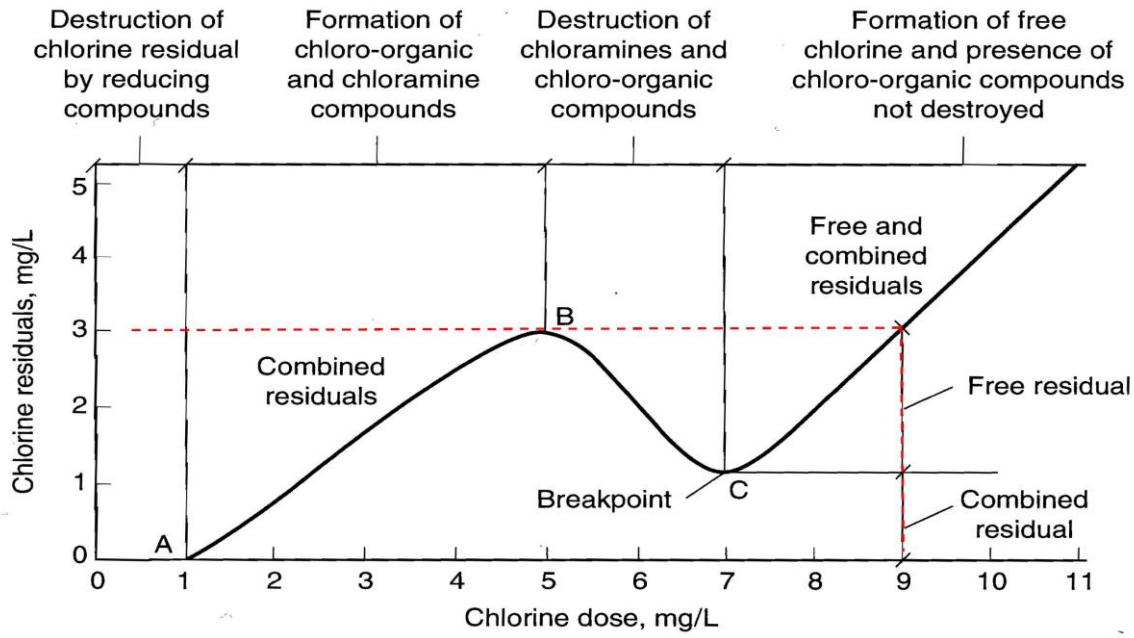
d) UV radiation

Chlorine demand & Breakpoint Chlorination

When chlorine is added to water, the chlorine at first oxidizes inorganic compounds & some amount of chlorine will be consumed by organic matter present in water. Then some amount of chlorine will be used for formation & subsequent destruction of chloramines.

The amount of chlorine required to fulfill the above consumption requirements are called **chlorine demand**. The point at which the chlorine demand is satisfied is called the **breakpoint chlorination**.

The excess dose of chlorine after breakpoint, is available for disinfection which appears as residual chlorine. A residual chlorine of minimum 0.5mg/L is maintained in disinfection practices.



C) Special or Advanced processes

The purposes of advanced water treatment processes are-

i) To take water treated by standard water treatment processes & to improve it to an exceptionally high-quality water as often required by particular industries like food industry & pharmaceuticals etc.

ii) To treat water containing specific chemical or microbiological contaminants, so that they can be brought to an acceptable standard.

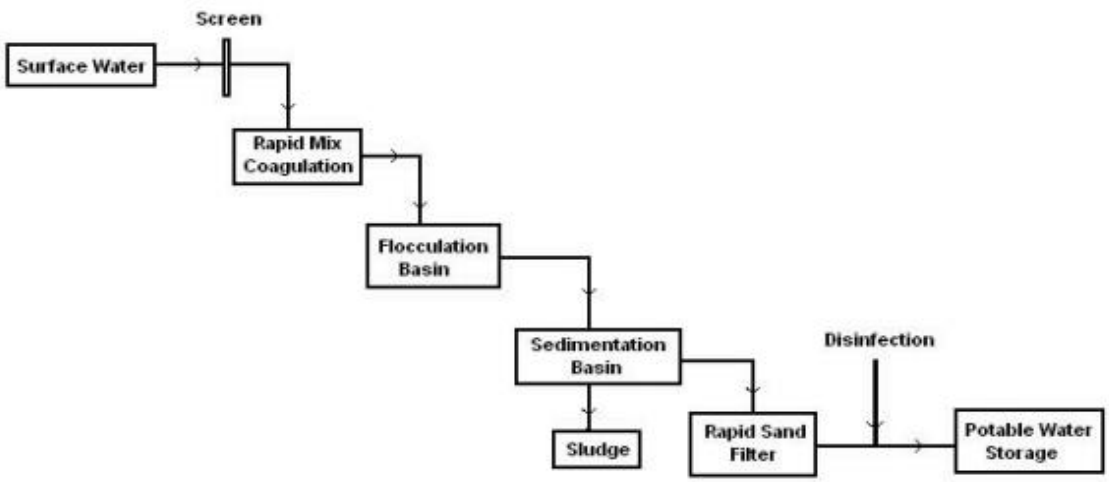
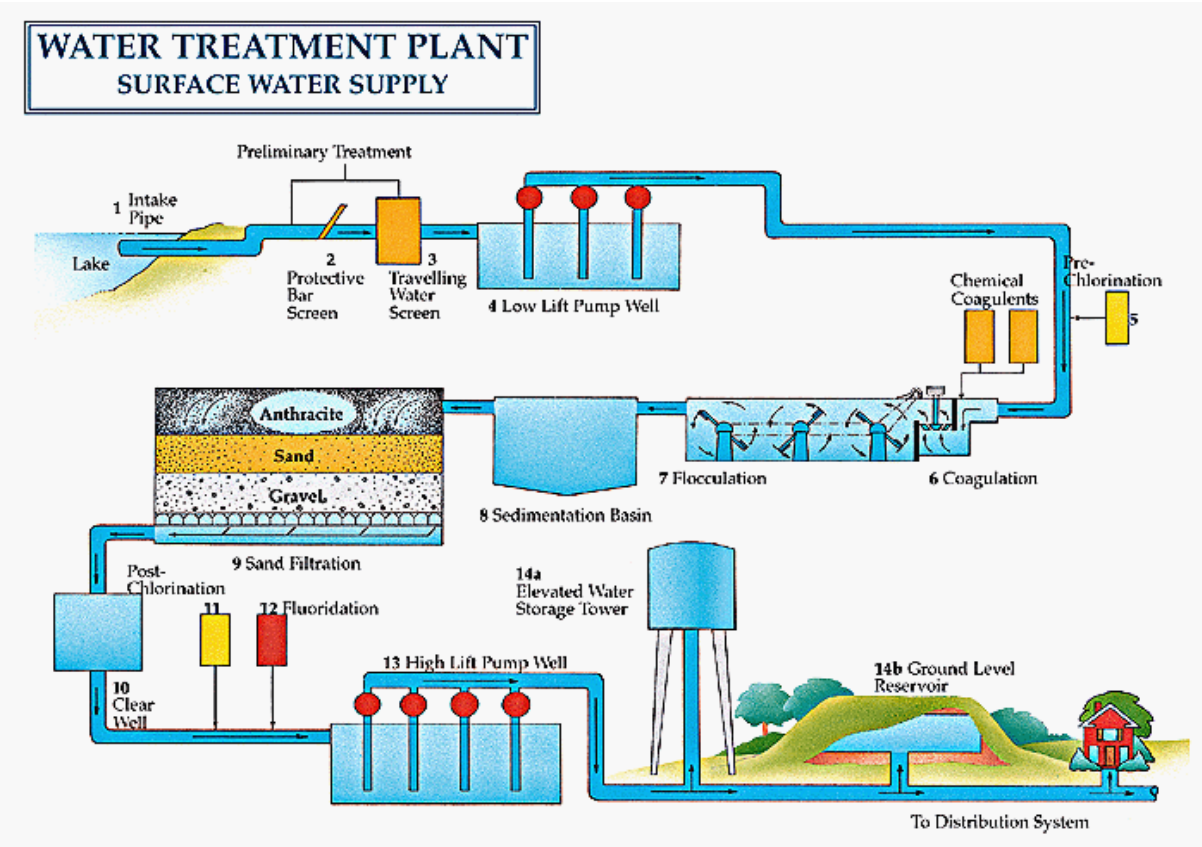


Figure 1 Flow diagram of a conventional potable water treatment plant.



Wastewater Treatment

Introduction

Generation of wastewater is based on our water consumption rate, which is increasing & the way we use water.

As undesirable contaminants are likely to be present in the wastewater, it can contaminate or degrade our surface water or ground water resources, hence it should be properly treated before releasing to the water bodies or environment.

Types of wastewater

a) Domestic/Sanitary wastewater

b) Industrial wastewater

c) Agricultural wastewater

d) Storm/Runoff water

Common Terms

Sewage- It refers to domestic / sanitary wastewater.

Sewer- It refers to pipes carrying sewage.

Sewerage- It refers to sewer network systems.

FOG- It refers to Fat, Oil & Grease, which are basically immiscible layer in the wastewater.

Wastewater Treatment Processes

a) Pre-treatment

b) Primary treatment

c) Secondary or Biological treatment

a) Pre-treatment

Pre-treatment of municipal wastewater is required to make it useful for further treatment & to decrease overall load on the system. It includes Screening, Grit Channels, Oil & Grease removal, Chemical pre-treatment etc.

b) Primary treatment

It basically removes suspended solids (SS) from the wastewater. Sometimes, it also removes some organic solids along with inorganic suspended solids. It includes sedimentation, coagulation etc. The removal efficiency w.r.t. SS is 50-70% and w.r.t. removal of BOD is 25-40%.

c) Secondary or Biological treatment

It can be divided into two major categories, depending on the availability of oxygen.

i) Aerobic Process i.e. it occurs in presence of oxygen &

ii) Anaerobic Process i.e. it occurs in absence of oxygen.

i) Aerobic Process

Aerobic Processes like Trickling Filter, RBC (Rotating Biological Contactor) & ASP (Activated Sludge Process) etc.

ii) Anaerobic Process

Anaerobic process has three major steps like Hydrolysis, Acidogenesis or Fermentation & Methanogenesis.

Aerobic vs Anaerobic Process

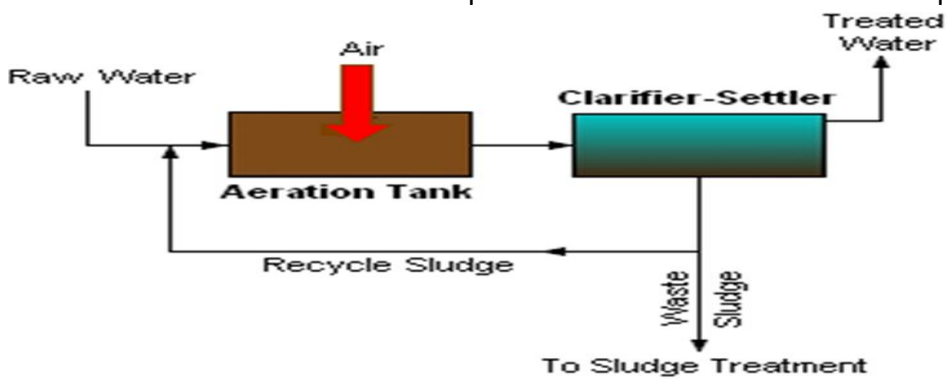
- In aerobic process, molecular oxygen present which is absent in case of anaerobic process.
- In aerobic process, stable oxidized end products like CO_2 & H_2O are generated, but in case of anaerobic process stable reduced end products like CH_4 & NH_3 are generated.

- In aerobic no byproduct recovery, but in anaerobic by product methane gas is useful.
- In aerobic appreciable microbial growth occurs, but in anaerobic no appreciable growth of microbes.
- In aerobic less skilled operation & in anaerobic more skilled/stringent operation is required.
- In aerobic low capital cost but high operating cost & in anaerobic high capital investment but low operating cost.
- Aerobic process is suitable for low BOD wastewater like sewage while anaerobic process is suitable for high BOD wastewater/ sludge coming from sugar mills and other agrochemical operations.

ASP (Activated Sludge Process)

It is an excellent method of treating either raw sewage or more generally the settled sewage. The sewage discharged from the primary sedimentation tank, which is normally used in this process, is mixed with 20 to 30% of activated sludge, which contains a large concentration of highly active aerobic micro-organisms, which increases the rate of decomposition.

- The ASP unit basically consists of CSTR (Continuously Stirred Tank Reactor), which is an aeration basin equipped with surface aerators and secondary clarifier unit.
- The aerobic bacteria culture is developed in the reactor and remain in suspension.



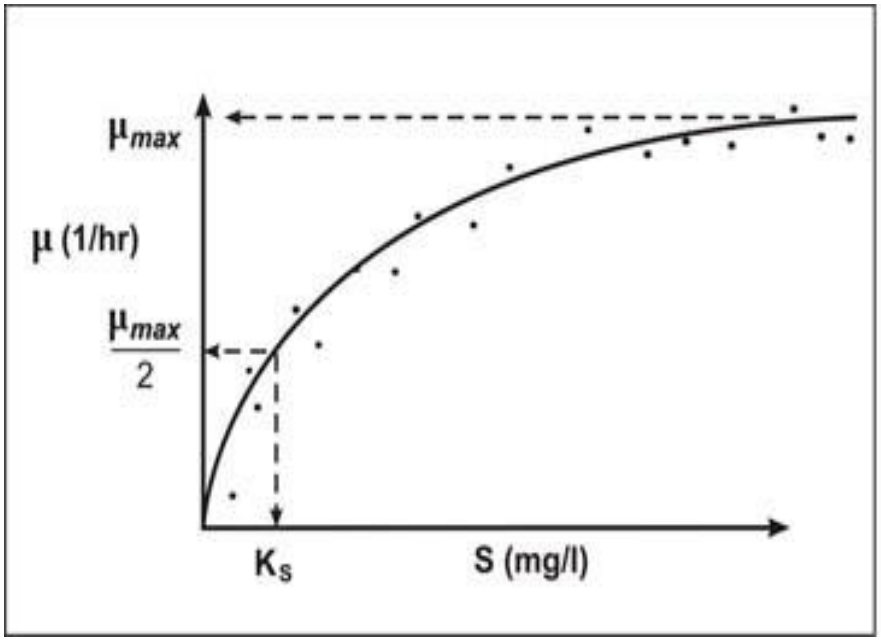
Monod Model

Monod Kinetics is used for microbial growth calculations and reactor sizing in the activated sludge process.

The pertinent equations are $r_g = dX/dt = \mu X = [\mu_{max}\{S/(K_s + S)\}]X$,

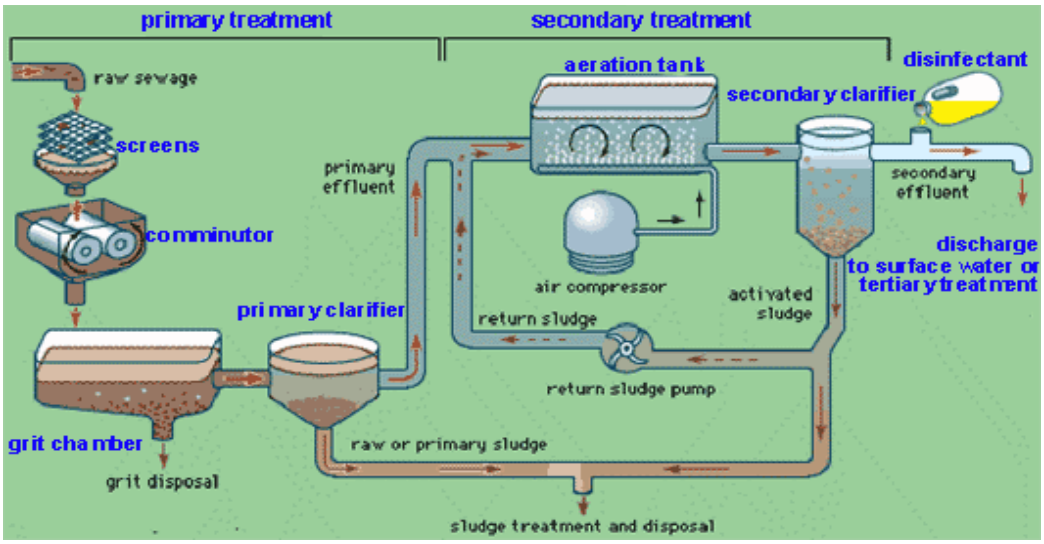
where the terms have the usual meanings. S, substrate Conc, X, microbe conc.

The diagram below shows how K_s is determined from μ_{max}. Monod's equation is an empirical equation to determine r_g, the growth rate of microbes.



$$\mu = \mu_{max}\left(\frac{S}{S+K_S}\right) \quad (5)$$

Wastewater treatment flow diagram



Reactor Configuration

In environmental engineering, particularly in water & waste water treatment reactors used are of 3 types –

i) Batch Reactor (BR)

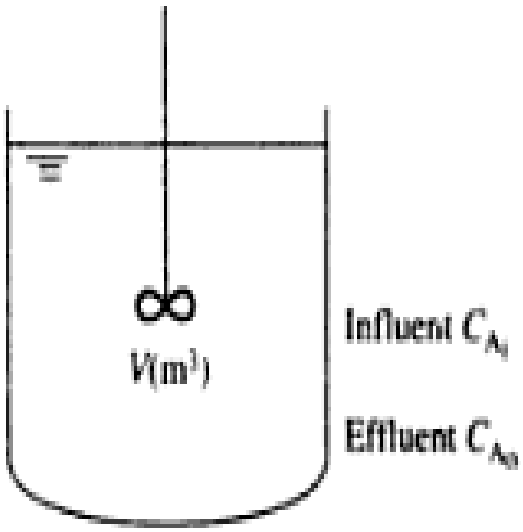
ii) Continuously Stirred Tank Reactor (CSTR)

iii) Plug Flow Reactor (PFR)

i) Batch Reactor (BR)

In this case, the reactants or inputs enter into the reactor, remain for desired time & then discharged.

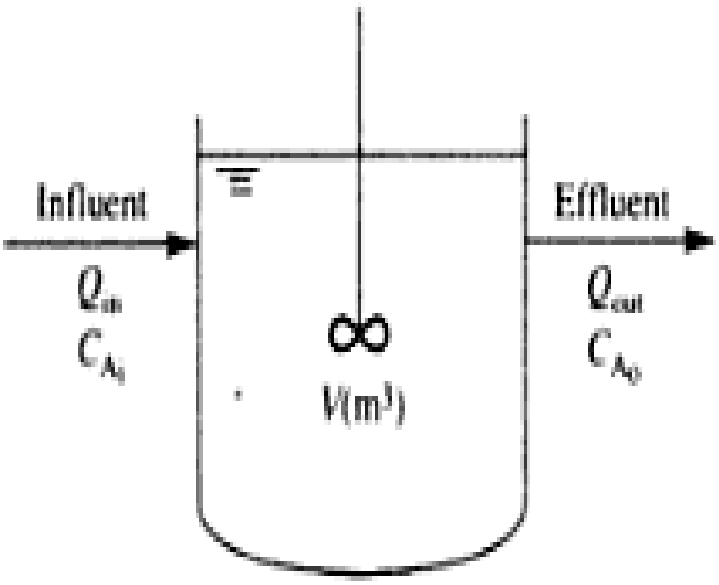
Ex:- BOD test



ii) Continuously Stirred Tank Reactor (CSTR)

In this case, reactants continuously enter into the reactor & the products are continuously discharged from a well-mixed vessel.

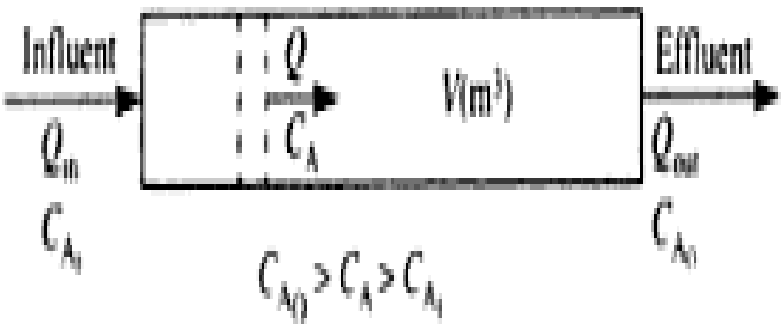
Ex:- Waste water treatment



iii) Plug Flow Reactor (PFR)

In this case, the reactants or inputs are entered at one end & products are discharged at the other end, after spending a minimum retention time in the system.

Ex:- mixing of pollutant in river flow



Module – 3: ATMOSPHERIC CHEMISTRY, SOIL CHEMISTRY, GROUND WATER RECHARGE, NOISE SOURCE & ABATEMENT

Soil Chemistry

The order of 10 most abundant elements in soil & crustal rocks are –

In soil:

O > Si > Al > Fe > C > Ca > K > Na > Mg > Ti

In Crustal rock:

O > Si > Al > Fe > Ca > Mg = Na > K > Ti > P

Soil minerals

The most common soil minerals are –

Quartz - SiO_2

Calcite - $CaCO_3$

Gypsum - $CaSO_4.2H_2O$

Gibbsite - $Al(OH)_3$

Soil Salinity

A soil solution is considered saline, if the electrical conductivity (EC) is greater than 4000 micro-siemens/cm or µs/cm. This condition occurs, when evaporation exceeds precipitation. If this condition continues then, saltpan will occur.

General Composition of soil

1)Inorganic or Mineral matter–45% (Ca,Mg etc.)

2) Organic matter – 5% (Humus)

3) Soil water – 25%

4) Soil air – 25%

Types of soil in India

1. Red soil
2. Black soil
3. Mountainous soil
4. Desert soil
5. Alluvial soil (fine grained fertile soil present in river bed)

Elemental properties of soil in relation to infiltration

1. **Bulk Density or Dry Density (ρ_b)**

   = M_d/V_t = Dry mass of soil/Total volume of soil

2. **Particle Density(ρ_m)** = M_d/V_d = Dry mass of soil/Dry volume of soil

3. **Porosity(ϕ)**= $(V_a+V_w)/V_s$ = (Volume of air + Volume of water)/Volume of soil

4. **The water content (Θ)** = V_w/V_s = Volume of water/ Volume of soil

5. **Degree of saturation(s)**: - It is a measure of wetness.

   So, $S = Vw/(Va+Vw) = (Vw/Vs)/[(Va+Vw)]/Vs = \Theta/\phi$

Soil Moisture Deficit (SMD)

The Soil Moisture Deficit is the term used when soil moisture is below the Field Capacity (FC) of soil.

Field Capacity (FC)

It is the maximum % of volumetric soil moisture that a soil sample will hold freely against earth gravity. It varies from 5% for sandy soil to 30% for dry soil.

Atmospheric Chemistry

Primary Pollutant

The Primary Pollutants are those which are emitted by identifiable manmade sources.

Ex:- SOX, NOX,CO

Secondary Pollutant

The secondary pollutants are those formed in the atmosphere by chemical or photochemical reactions of primary pollutants.

Ex:- Acid rain, Photochemical Smog

Photochemical smog

It is basically the product of HC(Hydrocarbon) + NOx + Sunlight

PAN

PAN is a class of compounds, Peroxy- Acyl- Nitrates, that are formed by photochemical reactions at the ground level. It is a key ingredient of SMOG, is a respiratory inhibitor and an eye irritant. The commonest PAN is Peroxy-Acetyl- Nitrate.

The reactions involving PAN formation are complex, and not totally well understood. However, the equations below give a rough pathway to the photochemical formation of PAN compounds.

PANs are secondary pollutants which are present in photochemical smog. PANs are not directly emitted as exhaust from power plants or internal combustion engines, but they are formed from other pollutants by chemical reactions in the atmosphere. Free radical reactions catalyzed by UV light from the sun oxidize unburned hydrocarbons to aldehydes, ketones & dicarbonyl compounds whose secondary reactions create peroxyacyl (the most common is the peroxyacetyl) radicals, which combine with the NO_2 radical to form Peroxyacylnitrates (PANs); if the radical is peroxyacetyl, then peroxyacetylnitrate (PAN) is formed.

The reactions involved, in a simplified manner, are as follows:

- NO_2 + hv -------> NO + O
- O + O_2 ---------> O_3
- O_3 + NO --------> NO_2 + O_2

When the nascent oxygen combines with hydrocarbon (i.e. HC or RC), then acyl radicals, RCO, are formed:

- RC + O ----------> RCO

Then these acyl radicals combine with oxygen molecule to form peroxy-acyl radicals

- RCO + O_2 -----> RCO_3

and finally, peroxy-acyl radicals RCO_3 combine with NO_2 to form RCO_3NO_2.

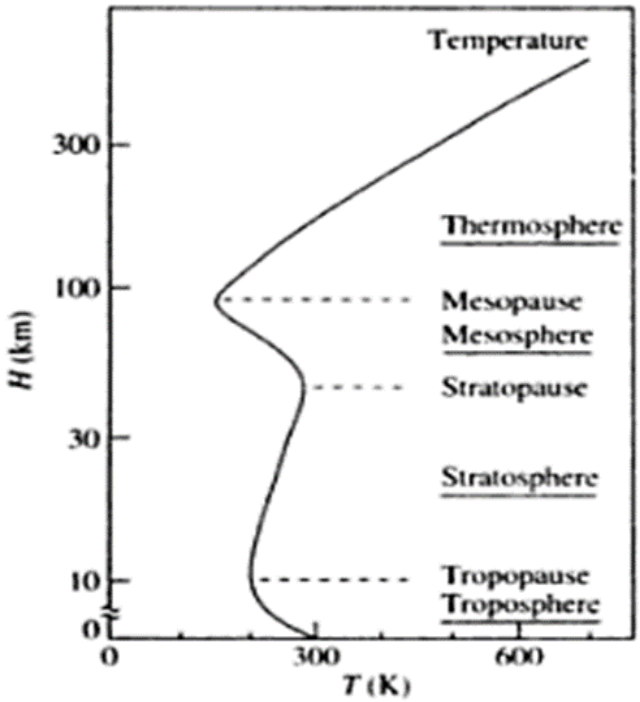
- RCO_3 + NO_2 ----------> RCO_3NO_2, a peroxy-acyl nitrate

If R is CH_3 (methyl group) then, $CH_3COOONO_2$ or $C_2H_3NO_5$ (peroxy-acetyl nitrate) is formed, which is called PAN.

Vertical Temperature Profile of earth's atmospheric Region

There are basically four regions are there.

- Troposphere – 0-12 km
- Stratosphere – 12-50 km
- Mesosphere – 50-80 km
- Thermosphere – 80-700 km



Vertical temperature profile of earth's atmospheric regions.

- The **Troposphere** is mostly heated by transfer of energy from the earth surface and temperature decreases with altitude or height.
- In **Stratosphere** temperature increases with height due to increased absorption of UV radiation by Ozone Layer.
- In **Mesosphere** temperature decreases with height due to decreasing solar heating.
- In **Thermosphere** temperature increases with height due to absorption of highly energetic solar radiation which causes atmosphere particles in this layer to become electrically charged.

Criteria Pollutants

The criteria pollutants as defined by USA, EC(European Community) & WHO includes CO,NO2,SO2,Pb,PM-10(Particulate Matter of dia less than 10µm). These are generally found in urban environment.

Non-Criteria Pollutants

The non-criteria air pollutants are those for which emission limits from industry are set or fixed. These are more poisonous & hazardous w.r.t. criteria pollutants.

Ex:- Benzene(C6H6),Carbon disulphide(CS2), Arsenic, Asbestos

Acid Deposition

Normal clean rain has a pH value of between 5.0 and 5.5, which is slightly acidic. However, when rain combines with sulfur dioxide or nitrogen oxides which are produced from power plants and automobiles the rain becomes much more acidic. Typical acid rain has a pH value of 4.0.

Hydrocarbon (HC), SO_X, NO_X emitted from industries & vehicles are the main source of acid deposition on the earth.

GHG (Green House Gases)

A GHG absorbs & emits radiation within the thermal infrared range having wavelength between 700nm to 1mm i.e. 700×10^{-9} m to 10^{-3}m. The major GHG in order of their contribution are CO2>CFCs>CH4>N2O(Nitrous Oxide)

Air Pollution Meteorology

The meteorological factors are –

- Wind speed & direction
- Temperature & humidity
- Turbulence
- Atmospheric Stability
- Topographic effects on meteorology

Scales

Air pollution emission can be measured in 3 scales-

a) Microscale

b) Mesoscale

c) Macroscale

a) Microscale

It is of the order of 1km & duration is minutes to hours.

Ex:- Chimney gases

b) Mesoscale

It is of the order of 100km & duration is hours to days.

Ex:- Mountain valley wind

c) Macroscale

It is of the order of 1000km & duration is days to weeks.

Ex:- wind over oceans & continents

Wind Speed

Wind speed at any height(z) can be determined using power law relationship i.e.

$$U_Z=U_{10}(Z/Z_{10})^P$$

Where,

UZ is wind speed at height Z mtr

U10 is wind speed at 10mtr height

Z is height in mtr

Z10 is 10 mtr height

P is exponent depends on terrain & stability class

Q. Use power law velocity profile equation to determine $U_{20}, U_{50}, U_{100}, U_{200}$ if U_{10} is 5m/sec & p is 0.2.

Lapse Rate

In the lower troposphere, the temperature usually decreases with altitude. The rate of temperature change or gradient is known as the lapse rate. Lapse rate can be ambient or adiabatic.

Ambient Lapse Rate (ALR) or Environmental Lapse Rate (ELR)

The temperature change or gradient with respect to rising altitude is known as ambient lapse rate & it varies from day to day & day to night. This is the actual change of temperature with altitude for the stationary atmosphere.

Adiabatic Lapse Rate

The temperature change of a parcel of air against rising altitude under adiabatic condition (i.e. occurring without the addition or loss of heat) is called the adiabatic lapse rate.

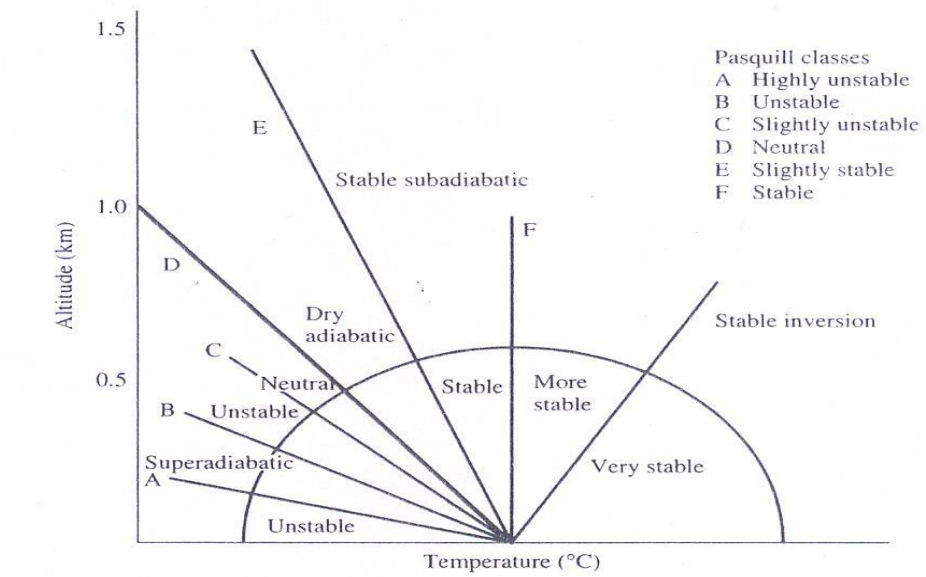
Adiabatic Lapse Rate can be of two types-

i) **Dry Adiabatic Lapse Rate(DALR)**
ii) **Moist adiabatic lapse rate (MALR) or Saturated adiabatic lapse rate(SALR)**

For dry air i.e. DALR value is 0.98^0C per 100mtr or approximately 1^0C per 100mtr or 10^0C per KM height.

For moist air i.e. MALR value is 0.55^{0}C per 100mtr or approximately 0.6^{0}C per 100mtr or 6^{0}C per KM height.

Pasquill – Gifford stability classes



Atmospheric Dispersion

For determining atmospheric dispersion, generally 5 models are used like –

- Gaussian model
- Numerical model
- Statistical model
- Emperical model
- Physical model

Out of the above models, **Gaussian model** is generally used.

The various assumptions of Gaussian model are –

i) There is no variation in wind speed & direction between the source & receptor.

ii) All discharges remain in the atmosphere.

iii) Dispersion doesn't occur in the downward direction.

iv) Emission rates are assumed constant & continuous.

Equipment Selection & Design

The Equipment Selection & Design is made on the basis of the types of compounds which are required to be removed. Generally, compounds can be of three types –

a) Volatile Organic Compounds (VOCs)

b) Inorganic Compounds

c) Particulate matter

The major types of equipments which are used are –

i) Incinerators

ii) Absorbers

iii) Adsorbers

iv) Condensors

v) Filters

vi) Scrubbers or wet collectors-

It is an apparatus using water or solution for purifying gases or vapours

vii) Various particle collection devices like ESP (Electrostatic Precipitator)

viii) Impingement separators-

By density difference, it separates solids from gases

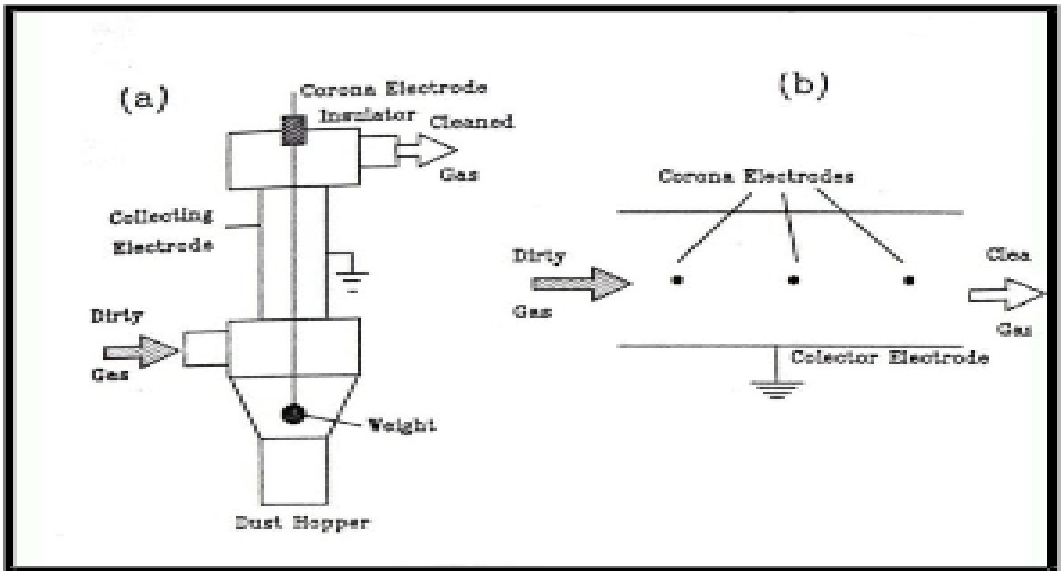
ESP (Electrostatic Precipitator)

An **electrostatic precipitator** (ESP) is a highly efficient filtration device that removes fine particles, like dust and smoke, from a flowing gas using the force of an induced electrostatic charge minimally impeding the flow of gases through the unit.

Basic Working Principle of ESP

In high- voltage electrostatic field, affected by the electric field force, gas ionization takes place. There are tremendous amount of electrons and ions existing in the ionized gas. After the dust particles are combined with these electrons and ions, they will be polarized, most of them are negatively polarized. Under the action of the field force, negatively charged particles migrates towards the positive electrode and in turn release electrons and attach to the positive electrode.

When the particles agglomerate and the layers reach a certain thickness on the plate, rapping system will start to work and the particles will be dislodged from the collecting plate by vibration and falling into the hopper. That ends the collection process.



The efficiency of an ESP with plate collector is given by Anderson-Deutsch equation which can be expressed as follows-

Efficiency = 1- $e^{(-AW/Q)}$

where,

A = area of the plates in square mtr

W= particle sedimentation velocity in an electric field in mtr/sec

Q= Gas flow rate in m^3/sec

Q. A quantity of 50 m^3/sec of air flows from a cement manufacturing facility. It contains cement particles whose settling velocity is 0.12 mtr/sec. If 99% removal efficiency is required, calculate the surface area of the ESP or design an ESP.

Q. Calculate the percentage increase in area of the ESP plates, if the efficiency of the ESP unit is to be increased from 99% to 99.7%. Given the flow rate is 50m^3/sec & particulate velocity is 0.15mtr/sec.

Q. The drift velocity of fly ash particle is given by the empirical equation w= $1.5X10^5$ d_p, where d_p is particle size. Determine the plate area required to remove particles of dia 0.7µm with removal efficiency 95%. Flow to ESP is 5m^3/sec.

Noise Pollution

Sounds are mechanical waves of pressure which allow us to hear & listen. Sounds can be transmitted through solid, liquid & gas. Generally human beings can perceive sound frequencies between 20 Hz & 20 KHz.

Noise is a sound, that is loud, unpleasant, unexpected or undesired.

Sounds, particularly loud ones, that disturb people or make it difficult to hear wanted sounds, are called noise. Noise is a subset of sound.

Sources of Noise Pollution

Sources can be classified into outside & inside. Outside include construction sites, road traffic, airport, factories etc. Inside include phones, tv, radio, home appliances etc.

Main source is urbanization & increasing population

Source	Sound in dB
Normal conversation	60
Alarm Clock	80
Car horn	120
Jet	130
Gunshot	140
Rocket Launching	180

Effects of Noise Pollution

- Sleep disturbance
- Lack of productivity
- Causes stress & depression
- Hormone imbalance
- Higher heart rate

Noise Control or Attenuation

Noise has following medium.

Source → Transmission Path → Receiver

The acoustic treatment of above three can help in attenuation.

Ways to reduce Noise Pollution

- Wear ear plugs
- Avoid using multiple appliances at a time
- Keep vehicles & machines in proper condition
- Buildings can be designed with suitable noise absorbing material for the walls, windows, and ceilings.
- Social awareness programs should be taken up to educate the public about the causes and effects of noise pollution.

Physical Properties of Sound

The various physical properties of sound are –

- **Amplitude(A)**- Amplitude is maximum or minimum pressure.
- **Wavelength(λ)**- Wavelength is the distance between successive crest or trough.
- **Time Period (T or P)**- Time period is the time gap between successive crest or trough
- **Frequency(f)**- Frequency is the number of complete pressure variations or cycles per second.
- **Speed of sound(c)**- Speed of sound is 340m/sec.

Noise Criteria

The various noise criteria can be –

- L_{Aeq} (Equivalent Continuous Level)
- L_{AE} (Sound Exposure Level)
- L_{AN} (Sound Level Exceeded for N% of time)

Indian Standard for ambient noise level

Area code	Area	Daytime	Nighttime
A	Industrial Area	75dB	70dB
B	Commercial Area	65dB	55dB
C	Residential Area	55dB	45dB
D	Silence Zone	50dB	40dB

The above values are of L_{Aeq}.

Inverse Square Law of sound Propagation

It states that, the sound intensity(I) is inversely proportional to the square of the distance (r) it propagates.

$$I = W/4\pi r^2 \text{ i.e.}$$

$$I \alpha 1/r^2,$$

where,

W is Sound power of noise source in watt.

The sound power level is measured in decibel(dB) i.e.

$L_W=10 \log(W/10^{-12})$

where,

L_W is sound power level in dB for 10^{-12} watt & W is sound power of noise source in watt.

The sound pressure level is measured in decibel(dB) & is directly proportional to the square of sound pressure i.e.

$L_P=10\log\{P^2/(P_0)^2\}$

$= 20 \log (P/P_0)$

Where,

L_P is sound pressure level in dB,

P is measured pressure in Pascal &

P_0 is reference pressure = 20µPa

Q. The sound from a voice shouting source is 0.001W. What is the sound power level?

Q.If a sound source has a pressure of 2000µPa at 10mtr distance, calculate

a) Sound Pressure Level in dB

b) Sound Intensity in Watt/m^2

c) Sound Power in Watt.

Q.If two sound sources have equal pressure of 2000µPa, then determine the sound pressure in dB.

Q. An air conditioner generates a noise level of 75dB for five minutes in every hour. If the background noise level is 55dB, then calculate the L_{Aeq}.

Module – 4: SOLID WASTE, HAZARDOUS WASTE & E-WASTE MANAGEMENT

Solid Waste Management

Solid waste is the unwanted or useless solid materials generated from combined residential, industrial and commercial activities in a given area.

Management of solid waste reduces or eliminates adverse impacts on the environment and human health and supports economic development and improved quality of life.

It may be categorised according to its origin (domestic, industrial, commercial, construction or institutional); according to its contents (organic materials like plastics, rubber, food items & inorganic materials like glass, metal etc); or according to hazard potential (toxic, non-toxic, flammable, radioactive, infectious etc).

A number of processes are involved in effectively managing waste for a municipality.

- These include monitoring, collection, transport, processing, recycling and disposal.

The order or hierarchy of solid waste management includes –

i) Waste prevention & minimization

ii) Reuse & Recycling

iii) Transformation

iv) Landfill

Reuse

A newly purchased product is put to another use after the first use is completed.

Recycling

It is the processing of used materials or waste into new products to prevent waste of potentially useful materials.

Properties of MSW (Municipal Solid Waste)

The various properties of MSW includes –

i) Physical Properties of MSW

ii) Biological Properties of MSW

iii) Chemical & Energy Properties of MSW

i) Physical Properties of MSW

The various Physical Properties of MSW are –

a) Particle Size Distribution

b) Density & Moisture Content

c) Field Capacity

d) Shear Strength

e) Hydraulic Conductivity

a) Particle Size Distribution

Particle size is measured in terms of size of screens in mm through which wastes are passing. This distribution will provide the information of various materials present.

b) Density & Moisture Content

Density is mass per unit volume, which is a useful physical parameter used for separating various wastes from each other before treatment.

The moisture content is expressed as mass of water vapour per unit mass of substance.

Moisture Content (%) = {(a-b)/a}X100

Where,

'a' is initial mass of sample &

'b' is mass of sample after drying

c) Field Capacity (FC)

It is the maximum percentage of volumetric soil moisture that a MSW sample will hold freely against earth's gravity.

FC can be calculated by –

FC = 0.6 – 0.55{W/(4500+W)}

Where,

W is overburden weight in Kg.

d) Shear Strength

Solid wastes when compacted usually have high shearing strength & hence don't flow on standing, but sludge has poor shearing strength & therefore very often sludge is co-disposed with MSW.

e) Hydraulic Conductivity

Sludge in landfills tend to resist the movement of water through them due to low hydraulic conductivity, as sludge has high moisture content.

ii) Biological Properties of MSW

Organic or biological matter in MSW is significant for the energy recovery by biodegradation.

Biodegradation can be accomplished either by aerobic or anaerobic process.

The biodegradability of the organic fraction of MSW is given by –

BF = 0.83 - 0.028LC

Where,

BF is Biodegradable Fraction

LC is Lignin(organic) Content in % of dry weight

High LC will give low biodegradability.

iii) Chemical & Energy Properties of MSW

The various steps involved in this are –

a) Proximate Analysis

b) Ultimate Analysis

c) Energy Content

a) Proximate Analysis

It deals with the determination of moisture content(W%), volatile matter(VM%), non-combustible fraction (i.e. ash%) & Fixed Carbon(FC).

FC can be found out by –

FC=100 - W(%) - VM(%) - ash(%)

FC is the solid combustible residue that remains after the material is heated & the volatile matter is ejected.

b) Ultimate Analysis

It is the elemental analysis of essential major elements like C,H,O,N,P,S in percentage mass.

c) Energy Content

Heating value or calorific value of MSW may be defined as the amount of heat liberated in calorie, when a gram of MSW is burnt. It gives the amount of organic matter present in the MSW.

Generally, terms like E, H_u, H_{wf}, H_{awf} are used to express energy content.

E is Energy Content

H_u is LHV(Lower Heat Value) i.e. energy received from waste as collected from site

H_{wf} is NHV(Normal Heat Value) i.e. energy received from water-free waste

H_{awf} is HHV(Higher Heat Value) i.e. energy received from ash-water-free waste

$$\mathbf{H_u = H_{awf} \times B - 2.445 \times W}$$

Where,

B is flammable fraction or combustible component

W is moisture content fraction or water content

The energy content of MSW can be determined from following equations –

i) Dulong Equation

ii) Khan Equation

i) Dulong Equation

$$H_{awf} = 337(C) + 1419\{(H) - 0.125(O)\} + 93(S) + 23(N)$$

Where, C, H, O, S, N are the % by weight of each element present in the material.

ii) Khan Equation

$$E = 0.051\{F + 3.6(CP)\} + 0.352(PLR)$$

Where,

E = Energy content,

F = % of food material by weight

CP = % of Cardboard & Paper by weight

PLR = % of Plastic & Rubber by weight

Separation of MSW

The different major components of MSW must be separated from one & the other in order to have suitable management of MSW.

The component separation can be done at the household or at the industry i.e. at the source or at the transfer station or at the final destination, where mechanical sorting or separation is possible.

Storage & Transport of MSW

It depends on types of collection facility available & materials present, which can be –

i) Door Step Collection

ii) Regular Roadside Collection

iii) Dustbins at market places

iv) Community Recycle bin

Integrated Waste Management

The ideal integrated waste management plan might have following priorities –

i) Minimize all components of waste fraction.

ii) Recycle paper, cardboard, non-ferrous metals.

iii) Reuse plastics, ferrous metals, glass.

iv) Compost food fraction of MSW.

v) Incinerate the remaining food waste.

vi) Landfill the remaining after proper treatment.

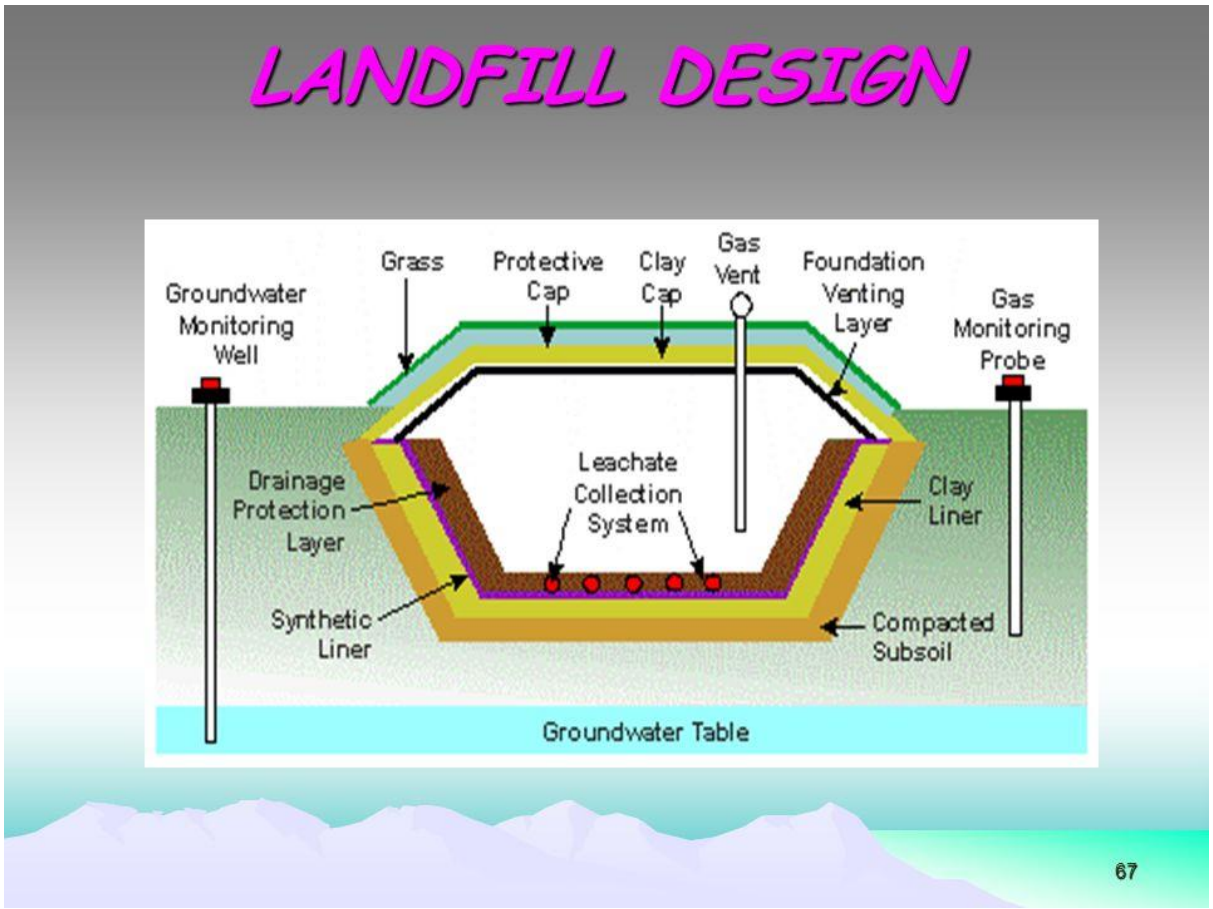
Land fill vs Open Dump

A Sanitary Landfill is a carefully engineered tract of land, designed by engineers. The system consists of a large excavated area in the ground, with a thick lining of plastic over which there is a compacted clay liner. The bottom of the pit also has a liner that helps in preventing the liquid waste (that comes out from the solid waste mostly if rainwater is soaked in the landfill) from leaking through, as it could contaminate the water supply. This liquid waste that gets collected is called leachate. The compacted garbage (or refuse) is then added to the landfill in the form of organized layers (layers of garbage alternating with layers of soil).

This is done for the elimination of any unpleasant odors and to expedite the rotting or decomposition process. When the landfill is fully covered, it is further sealed by a layer of compacted clay. Moreover, a gas (methane) is often produced as the waste deteriorates due to which a system is required to collect and pump it. If this is not managed effectively, the methane gas can explode. The methane collected can be utilized to generate electricity.

In most Western Countries, there are different types of landfills, mainly depending on the nature of waste materials. A landfill is built for each of the garbage types like household waste, hazardous chemicals or radioactive waste, construction waste, etc.

An open dump, as the name suggests, is an open area typically outside city limits where the garbage is dumped; this usually results in very unhygienic conditions and often pollutes the groundwater nearby, from toxic chemicals leaching into the groundwater.



Leachate in Landfills

Leachate is the contaminated water in landfills which arrive at the landfill site through external precipitation.

The amount of leachate produced in a landfill depends on it's water balance, which can be expressed as follows –

LC = PR + SRT – SRO – EP – ST

Where,

LC is Leachate,

PR is Precipitation

SRT is Surface Run To i.e. water outside the site entering the landfill

SRO is Surface Run-off

EP is Evapotranspiraton

ST is Change in water storage

Q. Calculate the Landfill area requirement for 20 years for a city of population of about 5 lakhs. Assume MSW generation as 500gm per capita per day & density of MSW is 500kg/m^3.

Hazardous Waste Management

Hazardous waste is defined as, any waste which because of it's physical, chemical quality, quantity and infectious characteristics can cause significant hazards to human health or the environment, when improperly treated, stored, transported or disposed.

A substance is hazardous, if it exhibits one or more of the following characteristics.

i) **Ignitable**:- The substance causes or enhances fire.

ii) **Reactive**:- The substance reacts with other materials & may explode.

iii) **Corrosive**:- The substance destroys tissues or metals.

iv) **Toxic**:- The substance is a danger to health, water, food & air.

Hazardous Category	Nature of Waste
H1	Explosive
H2	Oxidiser
H3A	Highly Flammable
H3B	Flammable
H4	Irritant
H5	Harmful

Generation of Hazardous Waste

Industry Type	Hazardous Substances
Battery	Cd, Pb, Ag, Zn
Electroplating	Co,Cr,Cu,Zn
Printing	As,Cr, Cu,Pb,Se
Textiles	Cr, Cu, Organics

Medical Hazardous Wastes

These are the wastes generated from hospitals, medical colleges, nursing homes, clinical laboratories & operation theatres etc.

Medical hazardous wastes include –

i) Expired or Obsolete medicines

ii) Infectious dressing materials

iii) Pathological wastes from medical laboratories & operation theatres

iv) Wastes from dental clinics etc.

Household Hazardous Waste

The household hazardous wastes come from kitchen, bathroom, garage, garden etc.

It may include –

i)Used or Exhausted batteries

ii)Chemicals like dyes, perfumes

iii)Pesticides, herbicides, used oils, lubricants etc.

Transportation of Hazardous Waste

The data or information required for Transportation of Hazardous Waste are –

i) Waste Generator

ii) Composition of Waste

iii) Physical Appearance

iv) Method of Packaging

v) ADR/RID classification

ADR means transport of dangerous goods by road

RID means transport of dangerous goods by rail

vi) UN Number(United Nation Number)

ADR/RID classification

Class	Item Description
1a	Explosive Items

1b	Items loaded with explosives
1c	Detonators & similar types of goods
2	Gases, Compressed, Condensed or dissolved under pressure
3	Flammable Liquids

UN Number (United Nation Number)

The UN numbers range from UN0001 to about UN3500 and are assigned by the United Nations Committee of Experts on the Transport of Dangerous Goods.

UN Number	Hazardous Item
1001	Acetylene Dissolved
1002	Compressed Air
1003	Liquefied air, refrigerated
1005	NH_3, dry, liquefied

Treatment of Hazardous waste

i) Thermal Treatment (Incineration)

ii) Chemical Treatment (Neutralization)

iii) Physical Treatment (Filtration, Flocculation)

iv) Disposal (Secure Landfill)

Incineration

Incineration can be defined as controlled high temperature oxidation of primarily organic compounds to produce CO_2 & H_2O with non-objectionable by-products.

3 T's of Incineration

i) Time

ii) Temperature

iii) Turbulence

i) Time

Adequate residence time for solids for complete destruction or breaking of bonds. It depends on feed rate, incinerator dimension & rotation speed.

ii) Temperature

High enough temperature is required for destruction of hazardous waste.

iii) Turbulence

Sufficient turbulence with adequate air or oxygen for mixing of waste with oxygen. This depends on rotation speed & incinerator types.

Types of Incinerator

i) Rotary Kiln

ii) Liquid injection

iii) Plasma Arc Destruction

iv) Wet Air Oxidation

V) Fluidized Bed Combustion

i) Rotary Kiln

This is most common type of incinerator. Here Chemical destruction of waste occurs.

ii) Liquid injection

This is for treating liquid organic waste.

iii) Plasma Arc Destruction

In this case, electric arc of very high temperature is used.

iv) Wet Air Oxidation

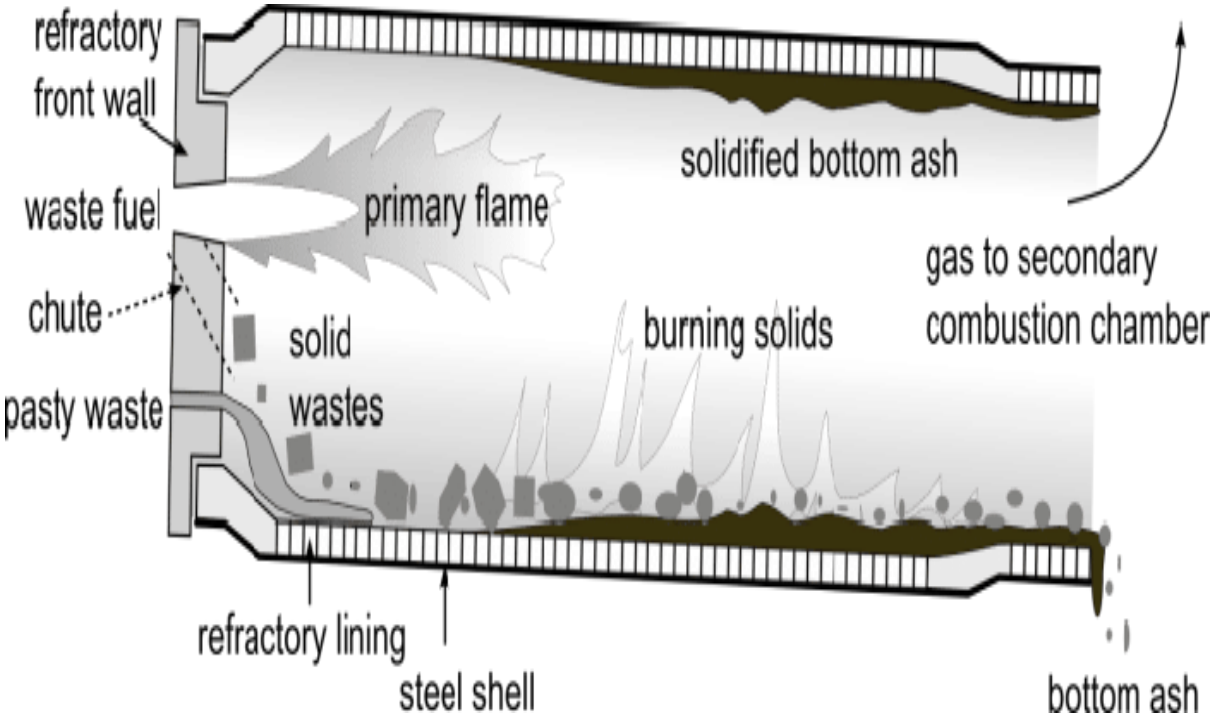
It is an aqueous phase oxidation where materials are exposed to gaseous source of oxygen.

v) Fluidized Bed Combustion

This is suitable for uniform types of waste. In this case, heat transfer is very fast & uniform.

Rotary kiln incinerator

A rotary kiln incinerator is typically a slightly inclined rotating cylinder, with dimensions around 2 m in diameter and 10 m long, on a pilot plant scale. There is a firebrick or refractory lining inside the cylinder. The kiln can handle all sorts of wastes and hazardous wastes including solids, liquids, sludges and gases. Partially combusted gases are often passed through a secondary combustion chamber for complete combustion. By the nature of its design and construction, the three Ts of combustion are adhered here - good contact or residence time, fair degree of turbulence, and operating at a high enough temperature. A schematic diagram is below :



DRE (Destruction & Removal Efficiency)

Destruction and removal efficiency (DRE) is the efficiency of the unit (kiln) in destruction and removal of a particular targeted organic compound.

In incineration DRE of above 99% is required.

DRE can be calculated by using the following formula –

DRE= $\{(W_{in} - W_{out})/W_{in}\} \times 100$

Where,

W_{in} is mass feed rate of specific organic component to the incinerator

W_{out} is mass emission rate of same organic component from the incinerator

CE(Combustion Efficiency)

Combustion efficiency is a measurement of how well the fuel being burned or is being utilized in the combustion process.

CE can be calculated by using the following formula –

CE = $\{(C_{CO2} - C_{CO})/C_{CO2}\} \times 100$

Where,

C_{CO2} is concentration of CO_2 in the emitted gas

C_{CO} is concentration of CO in the emitted gas

Q. Find out CE(Combustion Efficiency) & DRE(Destruction & Removal Efficiency) of an incinerator from the following data –

i) Input of phenolic waste is 18.2% & outlet phenol is 0.04%

ii) Outlet gas concentration of CO is 0.1% & CO_2 is 25.7%

E-Waste Management

E- waste or Electrical/Electronics waste is an environment killer. Today's electronic gadgets, tomorrow's electronic wastes.

'E-waste' means waste from electrical and electronic equipments; whole or in part or rejects from their manufacturing and repair process which are intended to be discarded as waste.

E-waste is the term used to cover all items of electrical and electronic equipment (EEE) and its parts that have been discarded by its owner as waste without the intent of re-use.

It is also referred to as WEEE (Waste Electrical and Electronic Equipment) or electronic waste or e-scrap in different regions.

Electronic waste or e-waste describes discarded electrical or electronic devices & used electronics which are destined for reuse, resale, recycling, or disposal are also considered e-waste.

Informal processing of e-waste in developing countries can lead to adverse human health effects and environmental pollution.

It is the fastest growing portion of municipal waste & its management will prevent -

- Pollution of ground water.
- Acidification of soil.
- Emission of toxic fumes and gases.
- Release of harmful gases into the air.

Sources/Types of E-Waste

1. Waste generated from the products used for data processing such as computers, computer devices like monitor, speakers, keyboards, printers etc.

2. Electronic devices used for entertainment like TV, DVDs, and CD players.

3. Equipment or devices used for communication like phones, landline phones, fax etc.

4. Household equipment's like vacuum cleaner, microwave ovens, washing machines, air conditioners etc.

5. Audio, visual components such as VCRs, Stereo equipment etc.

Components of E-Waste

EEE contains various materials including hazardous, valuable and scarce metals.

Common hazardous materials found in e-waste are heavy metals (such as mercury, lead, cadmium etc.)

E-Waste also contains many valuable materials (such as iron, copper, aluminum and plastics) and precious metals (like gold, silver, platinum and palladium) that can be recycled.

In fact, up to 60 elements from the periodic table can be found in complex electronics, and many of them are recoverable, though it is not always economic to do so presently.

From the resource perspective, e-waste is a potential "urban mine" that could provide a great amount of secondary resources for remanufacture and recycling.

E-Waste Composition (Approx.)

- Monitors - 10% ,
- Television -10%,
- Computers,Televisions, Fax, Printers etc.-15% ,
- DVD/VCR Player, CD Player, Radio etc.- 15% ,
- Refrigerators -20% ,
- Washing Machine, Dryers, AC, Vacuum Cleaners etc.- 30%

Is E-Waste hazardous?

E-Waste in itself is not hazardous. However, the hazardous constituents present in the e-waste render it hazardous & when such wastes are dismantled & processed, they pose hazard to health and environment.

Reasons of E-waste generation

- Advancement in technology.
- Changes in style, fashion and status
- Products nearing the end of their useful life
- Not taking precautions while using them

How E-Waste handled currently in India & across the world?

Around 90% of the e-waste generated in India are handled by the informal sectors. Incineration & Landfilling are used currently for majority of e waste.

Incineration i.e. E- Waste are destroyed by burning which produces harmful gases. Land filling which often leads to contaminate water and soil. Very less percentage of e-waste is recycled

RoHS and WEEE directives

- Companies selling electrical and electronic goods in the European Union must conform to the EU legislation for electrical and electronic equipment (EEE). The two most important directives are:
- The Waste Electrical and Electronic Equipment Directive (WEEE), which sets out the responsibilities of EEE producers for the collection and recycling of their products at the end of their lifecycle.
- The Restriction of Hazardous Substances Directive (RoHS), which bans the use of certain hazardous substances (such as lead, mercury, cadmium, hexavalent chromium and some polybrominated flame retardants) in EEE.
- WEEE and RoHS Directives are complex pieces of legislation that apply to a similar range of products.
- Each directive imposes obligations and outcomes that EU Member States must achieve. However, neither directive dictates how the Member States must fulfill their obligations. Therefore, when exporting to Europe, U.S. exporters need to be mindful that national rules for implementing each directive will differ from country to country.

Effects of e-waste on the environment

Emissions from E-Waste create environmental damage. Toxic chemicals from e-waste enter the "soil-crop- food pathway," and these are non-biodegradable and cause soil pollution. E-Waste dumping yards and nearby places are polluted and cause health hazards.

Effects of e-waste on human body

Elements present in e-waste affect human body in various ways.

Lead damage to central and peripheral nervous systems , blood systems and kidney damage and affects brain development of children.

Chromium leads to asthmatic bronchitis & DNA damage.

Cadmium leads to toxic irreversible effects on human health & accumulates in kidney and liver. causing neural damage

Mercury leads to chronic damage to brain and respiratory system.

Plastics including PVC - Burning produces dioxin which can cause reproductive and developmental problems, Immune system damage & Interfere with regulatory hormones etc.

Disposal of e-waste

We should ensure safe disposal of these e-waste, because –

1. Land fill disposal allows heavy metals to leach into ground water.

2. Incineration makes hazardous materials pollute air.

3. Disposal to water is dangerous and cause water and soil contamination.

Methods of Recycling

Disassembly/Dismantling

It is the systematic removal of components , parts or a group of parts or a subassembly from a product in e-waste.

Upgrading

It includes comminuting(breaking) and separation of materials using mechanical/physical or metallurgical processing .

Materials recovery

The materials are recovered by recycling facilities. The plastic, glass, metals can be recovered by sorting them before mixing them with other waste.

Advantages of recycling e-waste

- Assets recovery
- Reduction of need for landfills
- Resale and reuse
- Creation of jobs
- E waste has emerged as one of the fastest growing waste streams world wide today.
- Electronic gadgets, without proper disposal can cause environmental harm
- Reuse and Recycle are more beneficial than simple disposal.

How to Dispose E-waste

- Donate working older equipment to schools colleges or government entities in need.
- If computers are out of order then return it to the manufacturers. (HCL and Wipro in India has best take back service)
- Send waste goods to authorized recycling facility for proper disposal.

Probable Solutions

- Need for stringent health standards and environmental protection laws in India
- Extended producer responsibility
- Import of waste under license
- Producer-public-government cooperation
- Awareness program, following safer and efficient methods
- Choosing safer technology and cleaner substitute
- Monitoring of compliance rules

- Reduction of waste at source
- Recognition to the unorganized sectors in India.
- Designing of products using environment friendly raw items can make a change.
- Government should take steps to reduce e-waste disposal.

E- Waste (Management & Handling) Rules 2011

- E- Waste (Management & Handling) Rules 2011 were notified in 2011 and had come into force since 1st May, 2012.
- In order to ensure effective implementation of E-Waste Rules, Government of India modified the above one on 23.03.2016 which has been effective from 01-10-2016.
- These rules are applicable to every producer, consumer or bulk consumer, collection centre, dismantler and recycler of e-waste involved in the manufacture, sale, purchase and processing of electrical and electronic equipment or components
- Two categories of electrical and electronic equipment namely –
- (i) IT and Telecommunication Equipment and
- (ii) Consumer Electricals and Electronics such as TVs, Washing Machines, Refrigerators Air Conditioners including fluorescent and other mercury containing lamps are covered under these Rules.

Waste Minimization

Waste minimization is necessary to -

i) Control pollution

ii) Decrease the depletion rate of resources

iii) Increase efficiency & hence profitability

Elements of a waste minimization strategy

It includes –

i) Reduction at source

ii) Recycle/ Reuse

iii) Treatment

iv) Disposal

Waste Minimization Hierarchy

Waste Reduction Techniques

It has four major categories.

i) Inventory Management

ii) Production Process Modification

iii) Volume Reduction

iv) Recovery

i) Inventory Management

Inventory means detailed list of articles, goods, property.

Inventory management includes –

a) Inventory Control

b) Materials Control

ii) Production Process Modification

It includes –

a) Operational & maintenance procedures

b) Materials Change

c) Process equipment modification

iii) <u>Volume Reduction</u>

It includes –

a) Source Segregation

b) Concentration

iv) <u>Recovery</u>

It includes –

a) on-site

b) off-site

<u>Life Cycle Assessment (LCA)</u>

LCA is an useful environmental management tool.

Any product may have the following stages in it's lifecycle –

i) Raw material acquisition

ii) Bulk material processing

iii) Engineered & specialty materials production

iv) Manufacturing & assembly

v) Uses & services

vi) Retirement or expired

vii) Disposal

<u>Phases of LCA</u>

i) Planning

ii) Screening

iii) Data Collection

iv) Data treatment

v) Evaluation

Environmental Impact Assessment (EIA)

Environmental Impact Assessment (EIA) is a process that requires consideration of the environment & public participation in the decision-making process of project development.

Environmental Impact Assessment (EIA) is an important management tool for ensuring optimal use of natural resources for sustainable development.

Origin of EIA

All ecosystems including human beings have threshold of tolerance for pollution & disturbances, beyond which the system may suffer anything from temporary upsets to complete destruction. After second world war industrial & agricultural practices began to cause environmental damage which crossed the thresholds & lead to origin of EIA.

Stages of EIA

The stages of EIA include –

i) Screening

ii) Scoping

iii) EIS preparation

iv) Review

i) Screening

By screening, it is to be decided which projects should be subject to environmental assessment.

ii) Scoping

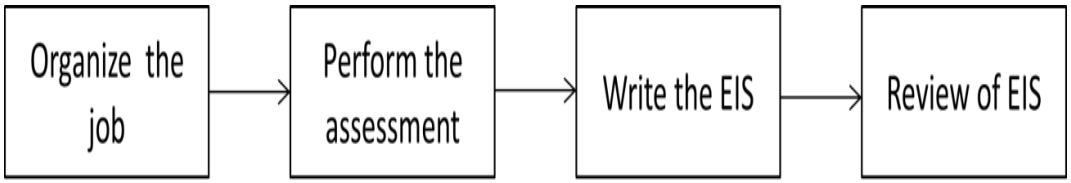
It is the process which defines the key or important issues that should be included in the environmental assessment.

iii) EIS preparation

It is the scientific & objective analysis of the scale, significance & importance of impacts identified.

iv) Review

The review panel guides the study & then advises the decision makers.



EIA Procedure

There are generally three options for establishing EIA procedures.

i) Legislative Option i.e. Legal approach

ii) Middle Ground Option i.e. within accepted planning & Procedures

iii) Policy Option i.e. within the administrative policy of government

Project Screening for EIA

The success of EIA depends on effective coverage & application of projects. The various methods commonly used to select projects for EIA are –

i) The use of positive (EIA required) & negative (EIA not required) list

ii) The use of project criteria

iii) The sensitive area criteria

iv) Matrices

v) Initial Environmental Evaluation(IEE)

Initial Environmental Evaluation (IEE)

IEE is a mini-EIA. It requires a description of the environment & the development & the identification of environmental impacts those are anticipated.

Environmental Impact Statement (EIS)

EIS is a review document prepared for assessment in the EIA process.

Scope Studies of EIS

Scoping is the procedure for establishing the TOR (Terms Of Reference) for the EIS. In general, the objectives would be to identify the concerns & issues those need attention, necessary for public involvement & to prepare a detailed report for the investigation of specific issues associated with the development.

Preparation of an EIS

One of the important elements of any project is the preparation of documentation to communicate the findings & conclusions of the study.

The value of the project is sharply diminished, if it's findings don't reach it's requisite results or intended audiences.

The important points to be considered are –

i) Planning

ii) Purpose

iii) Audience

iv) Structure

Review of an EIS

The functions of the review authority includes-

i) The scope of the assessment i.e. which projects should be subjected to a full or partial EIS.

ii) General or specific guidelines & advice on the methods of EIS.

iii) Formulate the TOR(Terms Of Reference) & initiate a detailed EIS.

iv) Ensure that, the EIS had been adequately completed within the TOR.

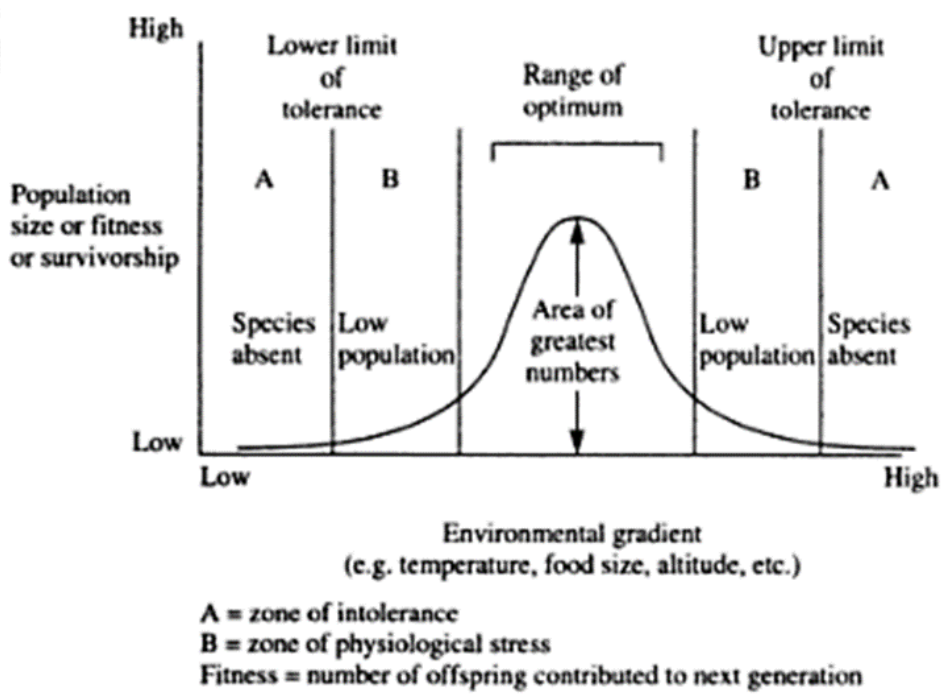
Module – 5: ENVIRONMENTAL GRADIENTS, LAWS AND HUMAN HEALTH

Environmental Gradients

The environment or environmental factors are not similar over the globe or around the earth, giving rise to Environmental Gradients & due to which different species live in different places.

The physical, chemical factors like light, temperature & P^H are known as **conditions** & the factors the organism actually uses like food, water, shelter are known as **resources**.

There is a global gradient change from the equator towards north or south.



A schematic tolerance curve for a single species population existing on a single environmental gradient.

This is a normally distributed bell-shaped curve, as the change in environmental conditions are gradual from equator towards north or south. This curve is best described by a Gaussian (Normal) Distribution function, of the form $f(x)=e^{-x2}$.

But, pollution can make it **skew-shaped** i.e. sharp decrease of curve towards left or right from range of optimum.

Limiting Factors

The resources in shortage of supply or the conditions over which the species has the smallest range of optimum will limit the species function & is called the limiting factor.

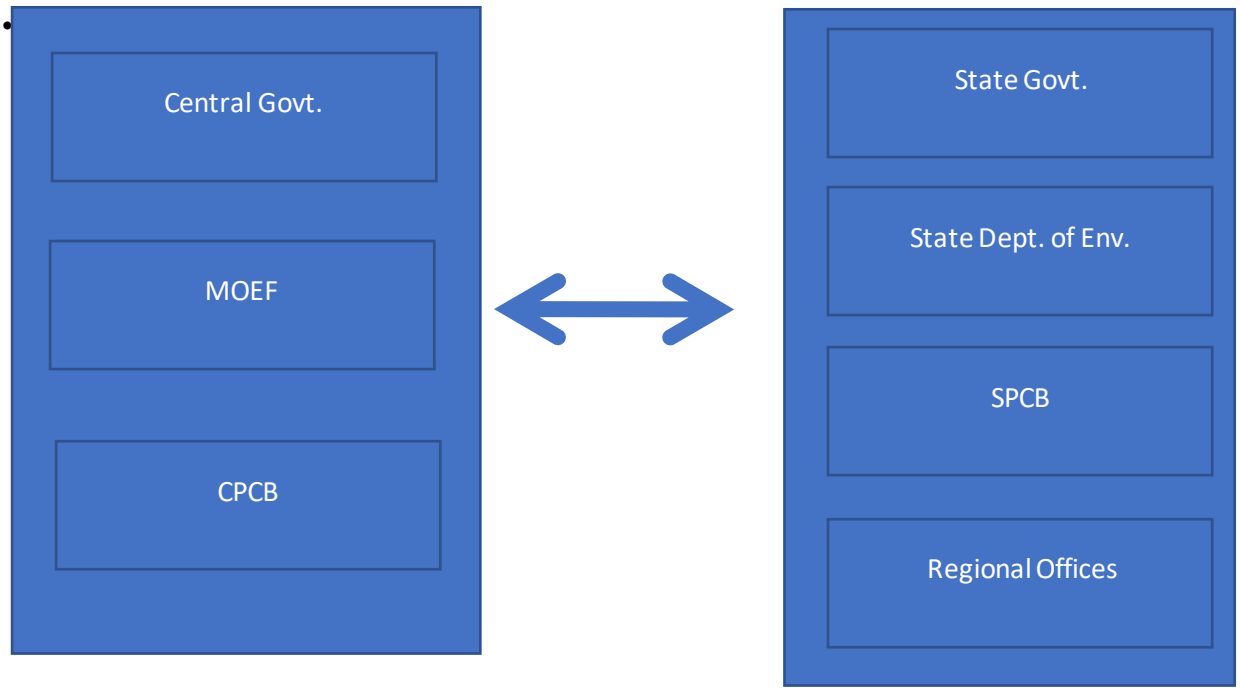
Ex:- Plant growth is strongly correlated with rain, but in arid region sufficient light is available, but water scarcity is there which is the limiting factor.

The tolerance of species to different environmental conditions or resources level vary from each other.

The **favourable** environmental factors to a species provide wide tolerance curve & broad range of optimum known as **Eurytopic** factors.

The **unfavourable** environmental factors to a species provide narrow tolerance curve & narrow range of optimum known as **Stenotopic** factors.

Regulatory Structure of Indian Env. Laws



MoEF – Ministry of Environment & Forest, Government of India

CPCB – Central Pollution Control Board

SPCB – State Pollution Control Board

Indian Env. Laws

- Water Laws
- Air Laws

- Wildlife & Forest Laws
- General Env. Laws

Water Laws

The various env. Laws on water are –

1) The Water (Prevention & Control of Pollution) Act – 1974

2) The Water (Prevention & Control of Pollution) Act rules – 1975

3) The Water (Prevention & Control of Pollution) Cess Act – 1977

4) The Water (Prevention & Control of Pollution) Act, rules & amendments – 1992

Objectives of Water Law

The objectives are –

a) Prevention & control of water pollution

b) Maintaining good quality of water

c) Establishment of boards for the prevention & control of water pollution.

Air Laws

The various env. Laws on air are –

1) The air (Prevention & Control of Pollution) Act – 1981

2)The air (Prevention & Control of Pollution) Act rules – 1982

3) Atomic Energy Act – 1982

4) Motor Vehicles Act – 1988

Objectives of Air Law

The objectives are –

a) Prevention & control of air pollution

b) Maintaining good quality of air

c) Establishment of boards for the prevention & control of air pollution

Wildlife & Forest Laws

The various env. Laws for wildlife & forest are -

1) The wildlife protection act – 1972

2) The wildlife protection act and amendments – 1991

3) The forest conservation act – 1980

4) The national forest policy – 1988

Objectives of Wildlife & Forest Laws

The Objectives are –

a) Protection & conservation of wildlife

b) To preserve biodiversity

c) To maintain essential ecological processes & life supporting systems.

General Env. Laws

The various laws in this category are –

1) The environmental protection act (EPA) – 1986 or Umbrella act

2) Hazardous waste (management & handling) rules – 1989

3) Biomedical waste (management & handling) rules – 1998

Objectives of Umbrella act

- Protection & improvement of environment & prevention of hazards to all living creatures.
- Maintenance of harmonious relationship between human beings & their environment.

National Green Tribunal (NGT)

The Indian govt. established a National Green Tribunal in October 2010, headed by a former judge of the supreme court. It is a tribunal exclusively dedicated to environmental issues. This body was established by an act of parliament i.e. National Green Tribunal Act of 2010, which will deal with all matters related to and arising out of environmental issues.

Eco-labelling of products

An eco-label is used to mark a consumer product to indicate that it is environment- friendly with regards to its manufacture and use. The label means that the product satisfies certain eco-friendly criteria laid down by the govt. In1991, the govt. of india launched its first eco-label called Eco-mark for easy identification of environment-friendly products.

Comprehensive Nuclear-Test-Ban Treaty (CTBT)

Comprehensive Nuclear-Test-Ban Treaty or CTBT bans nuclear explosive testing for everyone, everywhere, in all environments. CTBT makes very difficult for the countries to test or develop any nuclear weapon.

Some Important env. Laws

1. London Law of Sea (1962) is for prevention of sea pollution by oil.

2.Stockholm Agreement (1972) is for banning of ocean dumping wastes.

3. Montreal Protocol (1987) is for eliminating CFCs & other ozone depleting substances.

4. Rio Summit (1992) is with its objectives on sustainable development.

Kyoto Protocol

The Kyoto Protocol (1992) was signed to try and limit Greenhouse gases, based on the premise that Global Warming is occurring due to increased Greenhouse gas emissions.

Paris Agreement

The Paris Agreement of 2015 is to combat climate change and reduce the temperature rise to less than 2°C per century.

Environmental Institutions in India

1. TERI(Tata Energy Research Institute) established in1974.

2. NEERI(National Environment Engineering Research Institute) established in1958.

3. EPTRI(Environmental Protection Training and Research Institute) established in 1990.

4. IIFM(Indian Institute of Forest Management) established in 1982.

5. WII(Wildlife Institution of India) established in 1982 etc.

Environmental movements in India

1. Chipko movement started in 1973 to fight against deforestation.

2. Silent valley movement started against setting up of a 75MW hydroelectric project to protect species of flora and fauna.

3. Narmada Bachao Andolan started against a multipurpose project over river narmada by Baba Amte & Medha Patker.

4. Baliyapal Movement started against acquiring cultivated lands for testing missiles etc.

Important Env. Data

The idea of sustainable development was defined for the first time by the world commission on environment and development in the year-----

- **1987**

World Wildlife Fund was founded in ----

- **1961**

Earth day is observed on ----

- **April 22.**

World environment day is celebrated every year on -----

- **5th June**.

First International Conference on environment was held on ----

- **June,1972.**

Green Revolution

The Green Revolution within India commenced in the early 1960s that led to an increase in food grain production, especially in Punjab, Haryana, and Uttar Pradesh. Major milestones in this undertaking were the development of high-yielding varieties of wheat, and rust resistant strains of wheat.

Environment & Human Health

Env. Conditions Affecting Health

i) Cleanliness of working area

ii) Proper Lighting arrangement

iii) Sufficient Temperature & Ventilation

iv) Noise control

V) Dust control

Vi) Proper Working space & seating arrangements

Factors affecting Health

Physical factors like unhygienic living conditions, lack of education & overcrowding in small areas lead to health problems.

Chemical factors like improper or excess use of pesticides, herbicides and exposure to toxic, hazardous and radioactive substances lead to health problems.

Climate changing factors

Diseases of human and Their Control

Interaction of the agents like micro-organisms with the hosts like human beings and the environment can lead to various diseases.

Following **three factors** are important in this case.

- sufficient strength of micro-organisms,
- lower immunity of the person &
- favouring environment for the transmission of agents

Prevention can be done by -

- Enforcing immunization laws
- Providing health education to all
- Implementing good hand washing procedures.

Common Diseases

- Hepatitis is inflammation of the liver and often caused by viral infection.
- Typhoid is an acute illness associated with fever can be caused by bacteria salmonella typhi & spread to other people by contaminated water.
- Malaria is caused by a parasite called plasmodium which is transmitted to human beings by the bite of infected female anopheles species mosquitoes.
- Dengue fever is an infection caused by dengue viruses.

Role of IT in Env. & Human Health

- IT or Information Technology, is one of the fastest growing recent technologies & as technology is advancing, the role of IT is increasing day by day.
- IT is used in the development & application of computational tools to acquire, store, organize, achieve, analyze & visualize various data which can be used for observation & protection of env. & human health.

Role of IT in environment

- A large amount of information is easily available through remote sensing technology, GIS(Geographic Information System) & GPS(Global Positioning System) which can be used for various environmental purposes.
- This data and knowledge about global environment can be used for mapping & monitoring various natural resources, thereby helping in various environmental studies.

- GIS helps in detecting the existence of species of flora & fauna & counting the numbers of animals like tiger, elephant etc.
- MoEF(Ministry of Env. & Forest) & GOI(Govt. of India) have established ENVIS (Environmental Information System) in 1982 in different organizations for providing important information.

Role of IT in Human Health

- IT helps in the development of biotechnology, gene engineering, research & developments in drug technologies.
- IT plays a major role for a better human health in the fields like bioinformatics and online medical transcription.
- IT can be used for audio, visual & data communications for medical consultation, nursing & medical education.
- IT helps in spreading awareness about endemic, epidemic & communicable diseases.
- Using Remote sensing & GIS, areas which are prone to diseases like malaria, dengue etc. can be identified & accordingly steps for prevention or spreading of these diseases can be taken.

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